THIRSTY'S CODEX

STANDARDS EDITION

VOLUME X / 10 OF 20

PRODUCTION OPERATIONS AND COMMAND

Lifecycle Sections 301-332

NO PROOF, NO CLAIM

CONTINUOUS MASTER FOLIO / SOURCE-PRESERVING EDITION

Edition Statement

THIRSTY'S CODEX / STANDARDS EDITION / SOURCE-PRESERVING SET

This volume is one part of a continuous 20-volume edition. Canonical lifecycle section numbers remain stable across the set; no volume restarts or renumbers them.

The visual treatment has been rebuilt as a unified standards library: coordinated covers, volume architecture, source dividers, running headers, continuous master folios, local page numbers, and source-identity marks.

The source corpus remains authoritative over the presentation layer. Status terms are preserved exactly. CONTENT-COMPLETE does not silently become source-reconciled, hostile-reviewed, owner-accepted, ratified, or published.

Every registered source witness is identified by path, byte count, and SHA-256 digest. The complete source archive is preserved as an embedded attachment in Volume XX, while the two supplied foundational PDFs are preserved as attachments in Volume I.

CONTINUITY RULE

Read the master folio and canonical section sequence as one uninterrupted work. Each succeeding volume begins where the preceding volume ends.

Standards Edition Set Map

VOLUME I
Foundations and Governance
Doctrine, prompts, front matter, shared governance, roles, and output contracts

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Research, Strategy, and Risk
Lifecycle Sections 001-018

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Requirements through Hardware Architecture
Lifecycle Sections 019-060

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Canonical Authority and Assurance
Authority, constitutions, policies, schemas, provenance, evaluation, and verifica...

VOLUME XVI
Applied Corpus and V3Q
Agent-output corpus, V3Q implementation, fixtures, tests, and governed examples

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Training Views and Indexes
Continuous-pretraining, evaluation, retrieval, and training views with canonical ...

VOLUME XVIII
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Agents 101 and Owner's Cockpit source-standard facsimiles

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Static Analysis and Tooling Evidence
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301

SECTION 301 - SITE RELIABILITY ENGINEERING

CODEX/03_SECTIONS/301_SITE_RELIABILITY_ENGINEERING.txt

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5,046 BYTES / PRESERVED SOURCE IDENTITY

SECTION 301 — Site Reliability Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer and operate reliable services through service-level objectives, error budgets, automation, observability, incident learning, capacity, and sustainable human work.

REQUIRED OUTCOME

An SRE model with service ownership, user outcomes, indicators, objectives, budgets, alerts, runbooks, toil, incidents, reliability work, capacity, and recovery.

WHY THIS MATTERS

Reliability requires tradeoffs between change and stability, and between automation and human judgment. Uptime alone can hide correctness, latency, safety, privacy, support, or operator harm.

APPLICABILITY

Apply when a team owns ongoing service reliability, availability, performance, resilience, incident response, capacity, or operational improvement.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing service, operator, reliability, or operational outcome exists and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, service levels, observability, incident, capacity, performance, reliability, continuity, security, support, and staffing baselines.

REQUIRED INPUTS

Service, users, critical journeys, indicators, objectives, budget, dependencies, alerts, runbooks, incidents, capacity, toil, staffing, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Service owners define user outcomes, objectives, budgets, risk, on-call, change, incident, and staffing. Humans decide when to spend budget, pause change, accept risk, or protect operators.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate authorized telemetry, summarize incidents, draft runbooks, identify toil, and recommend bounded action. It must not suppress alerts, change production, close incidents, or accept reliability risk without authority.

APPLICABLE EXISTING AGENT ROLES

Site Reliability Engineer, Architect, Operations Engineer, Observer, Analyst, Incident Responder, Capacity Engineer, Tester, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define user outcomes and SLOs; instrument; alert; operate; respond; manage error budget; plan capacity; reduce toil; test failure and recovery; review changes; learn from incidents; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, journey, indicator, objective, budget, evidence, incident, action, authority, and uncertainty. Require human control for production and incident decisions and preserve on-call stop and escalation.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use meaningful indicators, multi-window alerts, runbooks, automation with guardrails, graceful degradation, redundancy, rollback, capacity, dependency visibility, and independent recovery.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show service health, user outcome, SLO, budget, incidents, dependencies, toil, capacity, evidence freshness, owner, and authorized pause, rollback, scale, escalate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

SRE charter; service and ownership map; SLI/SLO and budget; alert and runbook; capacity and toil; incident, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, journey, indicator, objective, period, source, method, result, uncertainty, owner, and limitation. Availability or dashboard health proves only the measured slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify indicators, objectives, alerts, runbooks, on-call, error budget, capacity, dependency, incident, recovery, automation, and operator workload. Exercise failures and false signals.

FAILURE STATES AND ANTI-PATTERNS

Uptime-only SLO, alert fatigue, budget ignored, toil hidden, automation without rollback, on-call burnout, incident blame, and service owner without authority.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, reduce change, fail over or roll back, communicate, protect operators, and reassess user outcome and SLO. Reopen on service, dependency, workload, or objective change.

COMPLETION CRITERIA

Service, user outcome, indicators, objectives, budget, owners, alerts, runbooks, incidents, capacity, evidence, workload, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve SLO history, runbooks, telemetry, incidents, decisions, evidence, exceptions, toil, capacity, and open gaps. Hand off to Production, Operations, Incident, Capacity, Reliability, Support, and Governance.

RELATED CODEX SECTIONS

029 Operational Architecture; 243-245 Reliability, Resilience, and Chaos Testing; 302 Production Engineering; 303-330 Observability, Incident, Capacity, Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, operations, reliability, incident, and owner-control context, and the integration roadmap. Exact services, users, SLOs, budgets, staffing, telemetry, and acceptance authority remain unresolved.

302

SECTION 302 - PRODUCTION ENGINEERING

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5,259 BYTES / PRESERVED SOURCE IDENTITY

SECTION 302 — Production Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Engineer production systems, workflows, controls, observability, capacity, release, incident, support, and recovery for safe and sustainable operation.

REQUIRED OUTCOME

A production-engineering model with service boundaries, environments, ownership, deployment, configuration, data, observability, performance, reliability, security, support, and recovery.

WHY THIS MATTERS

Production is where real users, data, commitments, and consequences meet. Production engineering turns design intent into controlled service behavior and must account for change, failure, people, and recovery.

APPLICABILITY

Apply when building, changing, operating, or improving a production or production-like service, system, environment, or workflow.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no production state, user, data, service, commitment, or operational control changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, release, deployment, configuration, platform, security, SRE, observability, incident, capacity, backup, and support baselines.

REQUIRED INPUTS

Service, users, environment, artifact, configuration, data, dependencies, SLO, telemetry, runbooks, staffing, support, incidents, capacity, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Production owners approve service, access, release, change, incident, communication, staffing, risk, recovery, and retirement. Humans retain control of consequential production action and user impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized production signals, summarize incidents, draft runbooks, identify drift, and recommend bounded actions. It must not change production, suppress alerts, access restricted data, communicate commitments, or close incidents without authority.

APPLICABLE EXISTING AGENT ROLES

Production Engineer, Site Reliability Engineer, Operations Engineer, Release Engineer, Security Auditor, Observer, Incident Responder, Capacity Engineer, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define production boundary and ownership; provision and secure; release; configure; observe; capacity-plan; support; respond; back up; recover; reduce toil; test; learn; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, environment, artifact, config, user impact, signal, action, authority, evidence, and uncertainty. Require human approval for production, data, security, recovery, and communication actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable artifacts, environment isolation, least privilege, progressive release, feature flags, telemetry, SLOs, graceful degradation, rollback, backup, restore, and independent control path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show production state, service health, changes, dependencies, incidents, capacity, cost, evidence freshness, owner, and authorized pause, rollback, scale, isolate, recover, communicate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Production charter; service and ownership map; release and change; configuration and environment; observability, support, capacity, incident, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, environment, version, config, data, user outcome, source, method, period, result, actor, and limitation. Production status or uptime proves neither correctness nor absence of harm.

VERIFICATION AND ACCEPTANCE TESTS

Verify deployment, configuration, security, observability, SLO, capacity, support, backup, restore, rollback, incident, communication, and retirement. Exercise dependency, operator, and control-plane failure.

FAILURE STATES AND ANTI-PATTERNS

Production as experiment without authority, alert suppression, shared admin, no rollback, support ignored, data recovery untested, and infrastructure health treated as user outcome.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, pause change, rollback or fail over, restore, communicate, support users, and reassess residual risk. Reopen on incident, service, dependency, data, owner, or commitment change.

COMPLETION CRITERIA

Service, environment, owners, changes, evidence, support, reliability, security, capacity, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve production baseline, runbooks, changes, telemetry, incidents, backups, decisions, evidence, exceptions, and open gaps. Hand off to SRE, Operations, Release, Support, Security, Data, and Governance.

RELATED CODEX SECTIONS

029 Operational Architecture; 267-268 Delivery and Deployment; 288-301 Release, Change, Deployment, DNS, CDN, and SRE; 303-330 Observability, Incident, Capacity, Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, production, operational, delivery, SRE, security, and owner-control context, and the integration roadmap. Exact services, environments, SLOs, staffing, data, recovery, and acceptance authority remain unresolved.

303

SECTION 303 - RUNTIME OPERATIONS

CODEX/03_SECTIONS/303_RUNTIME_OPERATIONS.txt

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4,904 BYTES / PRESERVED SOURCE IDENTITY

SECTION 303 — Runtime Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate live runtime processes, services, jobs, workloads, and control paths within their intended state, performance, security, support, and recovery boundaries.

REQUIRED OUTCOME

A runtime-operations model with process ownership, modes, health, configuration, dependencies, resource, scheduling, access, signals, change, incident, restart, recovery, and retirement.

WHY THIS MATTERS

Runtime state changes continuously and can diverge from design. A process can be running while serving bad data, leaking access, exhausting resources, or failing a critical user outcome.

APPLICABILITY

Apply when a live service, job, process, workload, device, agent, or control plane is executed, scheduled, restarted, paused, or recovered.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no runtime state, process, service, resource, dependency, or operating action changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, application, platform, configuration, deployment, observability, SRE, security, incident, capacity, and recovery baselines.

REQUIRED INPUTS

Runtime, version, config, process, schedule, identity, resources, dependencies, health, signals, runbook, SLO, incidents, support, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Runtime owners approve start, stop, restart, scale, change, access, incident, recovery, and retirement. Operators retain safe manual control; humans accept service and data risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized runtime signals, summarize state, draft diagnosis and runbooks, and recommend bounded actions. It must not restart, kill, scale, alter, or access restricted runtime state without authority.

APPLICABLE EXISTING AGENT ROLES

Runtime Engineer, Operations Engineer, Site Reliability Engineer, Application Engineer, Platform Engineer, Observer, Incident Responder, Tester, Verifier, and Security Auditor.

REQUIRED ACTIVITIES

Baseline runtime; start and schedule; observe; manage resources and configuration; handle health and dependencies; respond; restart or fail over; validate; document; reduce toil; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show process, version, environment, identity, signal, action, authority, evidence, and limitation. Require human approval for destructive, production, data, security, and recovery actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use health and business checks, graceful shutdown, idempotence, bounded retries, resource limits, dependency visibility, safe restart, state preservation, and control-plane independence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show runtime state, health, resources, dependencies, config, incidents, evidence freshness, owner, and authorized pause, restart, scale, isolate, recover, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Runtime charter; process and service catalog; runbooks; health and signal; incident, change, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify process, version, environment, identity, resource, signal, method, time, result, actor, and limitation. Running state proves neither correct behavior nor user outcome.

VERIFICATION AND ACCEPTANCE TESTS

Verify start, stop, restart, health, dependency, resource, access, configuration, logging, alert, recovery, support, and retirement. Test hung, duplicate, stale, and failed process.

FAILURE STATES AND ANTI-PATTERNS

Restart as diagnosis, unbounded retry, process health without business outcome, shared admin, unowned job, stale config, and runtime change without history.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, isolate or restart, restore state, fail over, communicate, and reassess user and data impact. Reopen on runtime, config, dependency, workload, or incident change.

COMPLETION CRITERIA

Runtime, owners, state, config, resources, dependencies, signals, runbooks, evidence, recovery, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve runtime baseline, schedules, configs, signals, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Application, Platform, SRE, Security, Incident, Capacity, and Support.

RELATED CODEX SECTIONS

039 Runtime Architecture; 258 Configuration; 290 Deployment; 301-302 SRE and Production; 313-317 Observability; 318-330 Incident, Capacity, Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, runtime, application, platform, SRE, and operations context, and the integration roadmap. Exact runtimes, signals, owners, SLOs, staffing, recovery, and acceptance authority remain unresolved.

304

SECTION 304 - CLOUD OPERATIONS

CODEX/03_SECTIONS/304_CLOUD_OPERATIONS.txt

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4,828 BYTES / PRESERVED SOURCE IDENTITY

SECTION 304 — Cloud Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate cloud accounts, services, resources, identities, data, networks, policies, costs, providers, and recovery within declared service and governance boundaries.

REQUIRED OUTCOME

A cloud-operations model with inventory, ownership, health, access, change, cost, quota, security, compliance, observability, incident, backup, recovery, and exit.

WHY THIS MATTERS

Cloud operations combine provider control planes, shared responsibility, dynamic resources, and billing. A healthy provider console can hide application, tenant, data, identity, or cost failure.

APPLICABILITY

Apply when cloud resources or managed services are provisioned, changed, monitored, secured, billed, recovered, or retired.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no cloud resource, account, provider, data, identity, cost, or operational behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use cloud architecture, IAM, security, data, configuration, infrastructure, cost, SRE, backup, recovery, and provider baselines.

REQUIRED INPUTS

Accounts, resources, regions, services, identity, policy, config, data, quotas, cost, telemetry, incidents, backup, recovery, and exit.

HUMAN RESPONSIBILITIES AND AUTHORITY

Cloud owners approve access, region, data, resource, cost, policy, outage, recovery, provider escalation, exception, and retirement. Humans accept shared responsibility and provider risk.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized cloud state, costs, metrics, and logs; draft runbooks and plans; and identify drift. It must not provision, modify, delete, access private data, or change policy without authority.

APPLICABLE EXISTING AGENT ROLES

Cloud Operations Engineer, Cloud Architect, FinOps Analyst, Security Auditor, Platform Engineer, Data Engineer, SRE, Tester, Verifier, and Incident Responder.

REQUIRED ACTIVITIES

Inventory and tag; secure accounts and identity; operate services; monitor; manage quota and cost; patch; respond; back up; restore or fail over; review provider changes; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show account, region, resource, service, identity, data, cost, action, authority, evidence, and limitation. Require human approval for privileged, destructive, data, cost, and regional actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use landing-zone controls, segmentation, least privilege, policy, budgets, quota, provider health, immutable artifacts, backup, restore, failover, and exit evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show cloud health, resources, quota, cost, data region, identity, incidents, evidence freshness, owner, and authorized scale, restrict, isolate, restore, fail over, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Cloud operations charter; account and resource catalog; access and policy; cost and quota; observability; incident, backup, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify account, region, resource, service, identity, config, provider, method, time, result, cost, owner, and limitation. Provider health or billing data proves only its view.

VERIFICATION AND ACCEPTANCE TESTS

Verify inventory, access, policy, quota, cost, health, data, backup, restore, failover, provider incident, audit, and exit. Test control-plane and region loss.

FAILURE STATES AND ANTI-PATTERNS

Cloud console as truth, admin everywhere, untagged resource, cost surprise, provider lock-in, shared account, backup untested, and provider status treated as application health.

RECOVERY AND RECONSIDERATION

Restrict account or resource, preserve state and logs, fail over or restore, rotate access, communicate, and reassess data and cost. Reopen on provider, region, service, resource, or policy change.

COMPLETION CRITERIA

Accounts, resources, owners, identity, policy, cost, health, evidence, backup, recovery, exit, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve cloud state, policies, metrics, costs, incidents, backups, decisions, evidence, exceptions, and open gaps. Hand off to Platform, Security, FinOps, Data, SRE, Support, and Governance.

RELATED CODEX SECTIONS

277 Cloud Engineering; 258 Configuration; 263 Environment; 301 SRE; 305 Network; 307 Data; 310 Platform; 323-330 Capacity, Availability, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, cloud, infrastructure, security, cost, data, and operations context, and the integration roadmap. Exact providers, resources, regions, cost, staffing, recovery, exit, and acceptance authority remain unresolved.

305

SECTION 305 - NETWORK OPERATIONS

CODEX/03_SECTIONS/305_NETWORK_OPERATIONS.txt

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4,670 BYTES / PRESERVED SOURCE IDENTITY

SECTION 305 — Network Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate network links, devices, routes, zones, DNS, access, policies, and connectivity while preserving service, security, evidence, and recovery.

REQUIRED OUTCOME

A network-operations model with inventory, health, configuration, change, access, capacity, incidents, provider, monitoring, failover, and runbook ownership.

WHY THIS MATTERS

Network operations handle high-blast-radius state. A route, firewall, DNS, or certificate change can silently redirect, isolate, expose, or degrade many systems.

APPLICABILITY

Apply when networks, links, devices, routes, policies, DNS, load balancers, wireless, or providers are monitored, changed, repaired, or recovered.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no network state, connectivity, trust, route, policy, or operational action changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use network architecture, identity, security, DNS, certificate, cloud, observability, change, incident, capacity, and recovery baselines.

REQUIRED INPUTS

Topology, devices, links, routes, zones, policies, identities, DNS, certificates, metrics, alerts, providers, changes, incidents, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Network owners approve access, route, firewall, DNS, maintenance, outage, failover, capture, exception, and recovery. Humans review high-blast-radius change.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized flow, telemetry, configuration, and incidents; draft diagnosis and runbooks; and recommend bounded actions. It must not change routes, ACLs, DNS, capture, or isolate without authority.

APPLICABLE EXISTING AGENT ROLES

Network Operations Engineer, Network Engineer, Security Auditor, Cloud Engineer, DNS Engineer, Observer, Tester, Verifier, and Incident Responder.

REQUIRED ACTIVITIES

Monitor; triage; maintain devices and links; change routes and policy; manage DNS and certificates; capacity-plan; fail over; troubleshoot; preserve evidence; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show device, link, route, zone, identity, action, authority, evidence, privacy boundary, and limitation. Require human approval for routing, firewall, capture, failover, and destructive action.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use management isolation, segmentation, authenticated control, redundant paths, configuration audit, flow visibility, rate controls, and out-of-band recovery.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show network health, topology, routes, zones, capacity, incidents, evidence freshness, owner, and authorized isolate, reroute, restore, fail over, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Network operations charter; inventory and topology; monitoring and alert; change and incident; capacity and failover; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify device, link, route, policy, zone, provider, method, time, result, actor, and limitation. Reachability or packet loss proves only the observed path and time.

VERIFICATION AND ACCEPTANCE TESTS

Verify monitoring, access, route, policy, DNS, certificate, capacity, alert, failover, rollback, provider incident, and evidence. Test stale state and control-plane loss.

FAILURE STATES AND ANTI-PATTERNS

Shared admin, unmonitored change, flat trust, route drift, capture without authority, single provider, alert fatigue, and network health equated with service health.

RECOVERY AND RECONSIDERATION

Isolate, preserve configs and flows, restore or reroute, rotate access, communicate, and reassess affected services and data. Reopen on topology, provider, threat, or policy change.

COMPLETION CRITERIA

Assets, topology, health, routes, policies, owners, evidence, capacity, failover, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve configs, topology, changes, incidents, flow evidence, decisions, backups, exceptions, and open gaps. Hand off to Cloud, DNS, Certificates, Security, SRE, Platform, and Operations.

RELATED CODEX SECTIONS

284 Network Engineering; 299 DNS; 298 Certificates; 300 CDN; 304 Cloud; 313-317 Observability; 323-330 Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, network, security, cloud, and operations context, and the integration roadmap. Exact topology, devices, providers, metrics, authority, privacy, recovery, and acceptance evidence remain unresolved.

306

SECTION 306 - SECURITY OPERATIONS

CODEX/03_SECTIONS/306_SECURITY_OPERATIONS.txt

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5,066 BYTES / PRESERVED SOURCE IDENTITY

SECTION 306 — Security Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Execute day-to-day security monitoring, control operation, triage, investigation, containment, access review, vulnerability response, evidence, and security improvement.

REQUIRED OUTCOME

A security-operations execution model with assets, signals, detections, operators, cases, playbooks, authority, privacy, evidence, escalation, containment, recovery, and measures.

WHY THIS MATTERS

Security operations translate controls into action. Unowned alerts, missing telemetry, excessive automation, poor evidence, and delayed response can let harm continue or create new harm through containment.

APPLICABILITY

Apply when security controls, alerts, identities, vulnerabilities, incidents, evidence, or defensive actions are operated continuously.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing security operation, signal, protected asset, response, or evidence obligation exists and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use security architecture, IAM, threat, vulnerability, observability, incident, privacy, legal, records, continuity, and staffing baselines.

REQUIRED INPUTS

Assets, telemetry, detections, alerts, cases, threats, vulnerabilities, identities, playbooks, staffing, privacy, retention, incidents, and response authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Security owners set severity, coverage, response, containment, evidence, notification, staffing, exceptions, and closure. Humans decide attribution, external communication, and material access or service action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate authorized signals, summarize cases, draft triage, and identify coverage or drift. It must not suppress, attribute, contain, revoke, delete evidence, access restricted data, or close cases without authority.

APPLICABLE EXISTING AGENT ROLES

Security Operations Analyst, Incident Responder, Threat Analyst, Vulnerability Analyst, Security Auditor, Observer, Analyst, Coordinator, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Collect and protect signals; triage; investigate; preserve evidence; review access; manage findings; contain; communicate; recover; tune; test playbooks; measure; and hand off incidents and lessons.

HUMAN-AI INTERACTION REQUIREMENTS

Show signal, source, asset, identity, time, confidence, action, authority, privacy, evidence, and limitation. Require human control for containment, attribution, notification, evidence, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate collection, detection, case, evidence, response, identity, retention, and control. Design for adversarial input, missing logs, privacy, clock drift, provider loss, and safe manual fallback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show coverage, alert, case, severity, evidence freshness, operator, authority, dependency, and authorized acknowledge, escalate, contain, release, preserve, communicate, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Security-operations charter; coverage and detection; triage and playbooks; case and evidence; vulnerability and incident; decision, exception, recovery, and continuity records.

EVIDENCE REQUIREMENTS

Identify asset, signal, source, clock, detection, actor, method, time, result, confidence, coverage, and limitation. Absence of alert proves only observed visibility.

VERIFICATION AND ACCEPTANCE TESTS

Exercise signal, false positive, missed event, log loss, tampering, alert volume, privacy, containment, evidence, recovery, communication, and closure. Verify operator response.

FAILURE STATES AND ANTI-PATTERNS

Alert fatigue, unowned playbook, attribution by score, auto-containment without authority, evidence loss, privacy overcollection, and security dashboard as proof of safety.

RECOVERY AND RECONSIDERATION

Preserve evidence, restore telemetry, narrow actions, contain safely, communicate, and reassess user and service impact. Reopen on threat, asset, signal, policy, staffing, or incident change.

COMPLETION CRITERIA

Assets, signals, detection, operators, authority, evidence, playbooks, response, recovery, privacy, measures, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve cases, detections, evidence, incidents, decisions, communications, exceptions, and open gaps. Hand off to Incident Response, Threat, Vulnerability, Blue Team, Privacy, Legal, Records, and Operations.

RELATED CODEX SECTIONS

206 Security Operations; 207-218 Threat, Vulnerability, Incident, Fraud, Abuse, Trust, and Privacy; 313-317 Observability; 323-330 Availability, Reliability, Continuity, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, security, observability, incident, privacy, and operations context, and the integration roadmap. Exact assets, tooling, staffing, signals, authority, retention, and acceptance evidence remain unresolved.

307

SECTION 307 - DATA OPERATIONS

CODEX/03_SECTIONS/307_DATA_OPERATIONS.txt

SHA-256 a009beb0b79900c243227d96fe66ec4677692f60b231f2508d97d26a4410c639

4,987 BYTES / PRESERVED SOURCE IDENTITY

SECTION 307 — Data Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate data pipelines, stores, quality, access, refresh, lineage, retention, deletion, recovery, and consumer services so data remains governed and useful.

REQUIRED OUTCOME

A data-operations model with sources, pipelines, stores, owners, schedules, quality, freshness, access, incidents, correction, backup, recovery, retention, and consumer communication.

WHY THIS MATTERS

Data operations failures propagate silently into reports, models, decisions, and obligations. A pipeline can be green while data is late, duplicated, unauthorized, or semantically wrong.

APPLICABILITY

Apply when data is ingested, transformed, stored, published, refreshed, accessed, corrected, archived, deleted, or recovered in ongoing operation.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no data source, pipeline, store, consumer, quality, access, lifecycle, or recovery behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use data architecture, quality, privacy, security, warehouse, lake, database, storage, records, AI, observability, and recovery baselines.

REQUIRED INPUTS

Sources, contracts, schemas, schedules, lineage, stores, consumers, quality, access, freshness, incidents, retention, deletion, backup, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Data owners approve meaning, access, refresh, correction, quarantine, retention, deletion, recovery, and consumer notice. Humans decide whether data is fit for consequential use.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized metadata, quality, lineage, and bounded samples; summarize incidents; and draft runbooks or remediation. It must not expose raw data, alter authoritative data, widen access, delete, or declare fitness without authority.

APPLICABLE EXISTING AGENT ROLES

Data Operations Engineer, Data Engineer, Data Quality Analyst, Data Architect, Privacy Reviewer, Security Auditor, Analyst, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Operate ingestion and transformation; monitor quality and freshness; manage access; reconcile; quarantine and correct; notify consumers; back up and restore; retain and delete; respond; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, schema, grain, time, quality, access, consumer, action, authority, evidence, and limitation. Require human approval for correction, quarantine, access, deletion, recovery, and consequential use.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use contracts, lineage, quality gates, quarantine, versioned pipelines, access control, privacy, freshness, reconciliation, backup, deletion propagation, and consumer impact.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show pipeline, source, freshness, quality, lineage, access, consumer, incidents, evidence freshness, owner, and authorized pause, quarantine, correct, backfill, restore, restrict, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Data-operations charter; source and pipeline catalog; quality and freshness; access and lineage; incident, correction, backup, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, pipeline, schema, grain, population, window, method, result, actor, quality, and limitation. Pipeline success or row count proves only execution and declared slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify ingestion, schema, quality, freshness, lineage, access, privacy, reconciliation, correction, backfill, backup, restore, deletion, and consumer notification.

FAILURE STATES AND ANTI-PATTERNS

Green pipeline, stale data, hidden correction, unrestricted raw zone, consumer unaware, deletion not propagated, backup not restored, and quality score as truth.

RECOVERY AND RECONSIDERATION

Quarantine, preserve versions, notify consumers, correct or backfill, restore, reconcile, and reassess downstream decisions. Reopen on source, schema, quality, pipeline, consumer, or lifecycle change.

COMPLETION CRITERIA

Sources, pipelines, stores, owners, quality, access, freshness, lineage, recovery, retention, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve data state, pipeline versions, incidents, corrections, backups, decisions, evidence, exceptions, and open gaps. Hand off to Data Engineering, Quality, Privacy, Security, Analytics, AI, Records, and Operations.

RELATED CODEX SECTIONS

031 Data Architecture; 159-162 Data Platforms; 218 Privacy; 254 Data Testing; 285-286 Database and Storage; 308 AI Operations; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data, privacy, quality, and operations context, and the integration roadmap. Exact sources, schedules, quality, consumers, authority, retention, recovery, and acceptance evidence remain unresolved.

308

SECTION 308 - AI OPERATIONS

CODEX/03_SECTIONS/308_AI_OPERATIONS.txt

SHA-256 ab3423a00a7d605588b840412223ddeef4a9c89d85ec1f4576e1aa4fecc58044

5,357 BYTES / PRESERVED SOURCE IDENTITY

SECTION 308 — AI Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate AI models, agents, prompts, tools, data, providers, policies, evaluations, human review, cost, safety, and recovery within explicit authority and use boundaries.

REQUIRED OUTCOME

An AI-operations model with identity, model and prompt versions, input and output data, tools, providers, policy, evaluation, monitoring, escalation, human control, rollback, and retirement.

WHY THIS MATTERS

AI behavior can drift with data, prompt, model, provider, tool, context, or policy changes. A successful request does not prove truth, safety, fairness, privacy, or suitability for a consequential decision.

APPLICABILITY

Apply when an AI model, agent, workflow, evaluator, retrieval system, tool call, or generated output operates for users, staff, systems, or decisions.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no AI model, agent, data, provider, prompt, tool, output, or human-impacting behavior is operated or changed and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use AI architecture, model testing, privacy, security, data, identity, policy, evaluation, incident, cost, platform, and operations baselines.

REQUIRED INPUTS

Model, provider, prompt, tool, data, policy, identity, use case, user, output, evaluation, latency, cost, safety, telemetry, fallback, and owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

AI owners approve intended use, model, provider, data, tools, authority, human review, thresholds, cost, incident, rollback, and retirement. Humans remain accountable for consequential decisions and user recourse.

AI RESPONSIBILITIES AND BOUNDARIES

AI may monitor authorized AI runs, identify drift or failures, summarize traces, and recommend bounded fallback. It must label generated content, preserve provenance, not self-authorize, access restricted data, conceal errors, or change model, prompt, policy, tool, or user authority without human control.

APPLICABLE EXISTING AGENT ROLES

AI Operations Engineer, AI Architect, Model Evaluator, Security Auditor, Privacy Reviewer, Data Engineer, SRE, Incident Responder, Tester, Verifier, and Governance Reviewer.

REQUIRED ACTIVITIES

Inventory model and provider; manage versions and prompts; secure data and tools; evaluate; deploy; monitor quality, safety, cost, and drift; review human escalation; respond; fallback; record; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show model, provider, prompt, data, tool, identity, output, confidence, human review, cost, evidence, and limitation. Require human approval for high-impact output, tool action, data access, policy, rollback, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Trace model, prompt, retrieval, tool, policy, input, output, reviewer, and action. Provide guardrails, sandbox, rate limits, redaction, fallback, provider substitution, evaluation, kill switch, and durable audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show AI service, model, provider, version, health, drift, safety, cost, human-review rate, incidents, evidence freshness, owner, and authorized pause, restrict, fallback, rollback, revoke, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

AI operations charter; model and provider registry; prompt and tool records; evaluation and monitoring; incident and fallback; human-review, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify model, provider, prompt, input, data, tool, policy, output, reviewer, method, time, result, and limitation. Fluency, confidence, or request success proves neither truth nor safety.

VERIFICATION AND ACCEPTANCE TESTS

Verify model and prompt version, data, privacy, security, tool authorization, output quality, safety, drift, cost, latency, human review, fallback, rollback, and incident handling.

FAILURE STATES AND ANTI-PATTERNS

AI output as fact, prompt drift, hidden provider change, tool overreach, no human escalation, sensitive data leakage, evaluation on narrow examples, and cost or latency unbounded.

RECOVERY AND RECONSIDERATION

Pause or narrow AI use, preserve traces, route to human or fallback, revoke tools, switch provider or model, correct affected decisions, notify owners, and reassess harm. Reopen on model, data, prompt, provider, policy, tool, or population change.

COMPLETION CRITERIA

Use, model, provider, prompt, tools, data, authority, evaluation, monitoring, human control, recovery, evidence, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve model and prompt versions, traces, evaluations, incidents, decisions, evidence, exceptions, cost, and open gaps. Hand off to AI Engineering, Data, Security, Privacy, SRE, Support, Governance, and Retirement.

RELATED CODEX SECTIONS

157-158 AI and ML Engineering; 200-205 IAM and Protection; 217-218 Trust and Privacy; 253 Model Testing; 301 SRE; 307 Data Operations; 369-371 Experimentation and Optimization.

CODEX/03_SECTIONS/308_AI_OPERATIONS.txt

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, AI, data, privacy, security, evaluation, and operations context, and the integration roadmap. Exact models, providers, data, tools, authority, metrics, human review, and acceptance evidence remain unresolved.

309

SECTION 309 - APPLICATION OPERATIONS

CODEX/03_SECTIONS/309_APPLICATION_OPERATIONS.txt

SHA-256 989fd1f253705ad55cd99f343f5165640be5a92e7565f40cd71775a1be3d2709

4,812 BYTES / PRESERVED SOURCE IDENTITY

SECTION 309 — Application Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate applications and their runtime dependencies so user outcomes, data, security, support, performance, and recovery remain within declared boundaries.

REQUIRED OUTCOME

An application-operations model with service owner, versions, configuration, dependencies, health, jobs, data, access, release, support, incidents, and recovery.

WHY THIS MATTERS

Application behavior is more than process health. Feature, data, dependency, user, support, and operational failure can occur while the application reports as running.

APPLICABILITY

Apply when an application or service is deployed, configured, monitored, supported, changed, recovered, or retired.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no application, service, user, data, configuration, dependency, or operational behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use application architecture, runtime, release, configuration, data, identity, security, observability, SRE, support, and recovery baselines.

REQUIRED INPUTS

Application, version, config, users, data, dependencies, health, jobs, metrics, logs, traces, incidents, support, SLO, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Application owners approve release, config, access, data, support, incident, recovery, exception, and retirement. Humans decide user-impact and service-risk tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized application signals, logs, traces, and incidents; draft diagnosis and runbooks; and recommend bounded action. It must not alter, access restricted data, suppress alerts, or close incidents without authority.

APPLICABLE EXISTING AGENT ROLES

Application Operations Engineer, Application Engineer, SRE, Operations Engineer, Security Auditor, Incident Responder, Tester, Verifier, and Support Reviewer.

REQUIRED ACTIVITIES

Operate runtime and jobs; monitor health and outcomes; manage config and releases; support users; respond; manage data and dependencies; back up; recover; reduce toil; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show app, version, config, user journey, data, signal, action, authority, evidence, and limitation. Require human approval for production, data, security, recovery, and communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use health plus business indicators, structured logs, traces, feature control, graceful degradation, idempotent jobs, rollback, data recovery, and support diagnostics.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show application state, version, dependencies, health, incidents, user impact, evidence freshness, owner, and authorized pause, rollback, scale, restrict, recover, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Application operations charter; service and dependency map; runbooks; release and config; monitoring and support; incident, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify app, version, environment, user outcome, data, signal, source, method, time, result, actor, and limitation. Process health proves neither application correctness nor user success.

VERIFICATION AND ACCEPTANCE TESTS

Verify release, config, access, data, health, job, dependency, alert, log, trace, support, backup, restore, rollback, and retirement. Test user-impacting failure.

FAILURE STATES AND ANTI-PATTERNS

Process-only monitoring, hidden job failure, config drift, log without context, support unprepared, recovery without data reconciliation, and application uptime treated as outcome.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, pause change, rollback or fail over, restore data, communicate, support users, and reassess. Reopen on app, dependency, data, config, user, or incident change.

COMPLETION CRITERIA

Application, users, data, versions, config, dependencies, signals, support, recovery, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve app baseline, runbooks, releases, configs, incidents, telemetry, backups, decisions, evidence, exceptions, and open gaps. Hand off to Platform, SRE, Security, Data, Support, and Release.

RELATED CODEX SECTIONS

033 Application Architecture; 258 Configuration; 290 Deployment; 301-303 SRE, Production, and Runtime; 313-317 Observability; 323-330 Availability and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, application, runtime, data, security, SRE, and operations context, and the integration roadmap. Exact applications, journeys, SLOs, telemetry, staffing, recovery, and acceptance authority remain unresolved.

310

SECTION 310 - PLATFORM OPERATIONS

CODEX/03_SECTIONS/310_PLATFORM_OPERATIONS.txt

SHA-256 e6ca6ba84eacd6bd164ba0c25f84bd5124de27a95d4ff8ceeb192dc806e373e9

4,944 BYTES / PRESERVED SOURCE IDENTITY

SECTION 310 — Platform Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate shared platforms, control planes, clusters, services, guardrails, self-service paths, resources, and support so consumer workloads remain secure, reliable, and governable.

REQUIRED OUTCOME

A platform-operations model with capabilities, consumers, ownership, health, access, capacity, cost, policy, change, support, incidents, recovery, and retirement.

WHY THIS MATTERS

Platform failure has multiplier effect. A control-plane outage, bad template, policy error, capacity issue, or hidden dependency can affect many teams and make local recovery impossible.

APPLICABILITY

Apply when a shared platform or service is provisioned, changed, monitored, supported, secured, recovered, or retired.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no shared platform, consumer, guardrail, resource, service, or operational behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use platform architecture, cloud, infrastructure, identity, security, configuration, DevOps, SRE, support, cost, and recovery baselines.

REQUIRED INPUTS

Platform capabilities, consumers, services, resources, policy, access, health, capacity, cost, incidents, support, dependencies, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Platform owners approve service, guardrails, access, change, cost, support, incident, recovery, exceptions, and retirement. Consumers retain workload responsibility and escalation rights.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized platform state, consumer signals, and incidents; draft runbooks; and recommend bounded operations. It must not change shared policy, grant access, provision, or hide consumer impact without authority.

APPLICABLE EXISTING AGENT ROLES

Platform Operations Engineer, Platform Engineer, SRE, Security Auditor, Cloud Engineer, Support Reviewer, Incident Responder, Tester, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Operate platform services; monitor health; manage access and capacity; support consumers; deploy and change; handle incidents; maintain guardrails; recover; communicate; and retire or migrate capabilities.

HUMAN-AI INTERACTION REQUIREMENTS

Show platform, capability, consumer, policy, resource, action, evidence, authority, and limitation. Require human approval for platform-wide, privileged, destructive, cost, and recovery actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use tenant isolation, policy, service boundaries, self-service guardrails, control-plane redundancy, observability, backup, rollback, escape hatch, and consumer impact visibility.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show platform health, consumers, resources, capacity, cost, policy, incidents, evidence freshness, owner, and authorized restrict, scale, rollback, recover, migrate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Platform operations charter; capability and consumer map; health and support; access, capacity, cost, incident, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify capability, platform, consumer, resource, policy, method, time, result, cost, owner, and limitation. Platform health proves neither consumer outcome nor workload correctness.

VERIFICATION AND ACCEPTANCE TESTS

Verify platform service, consumer isolation, access, guardrails, capacity, cost, support, observability, change, rollback, recovery, and provider loss.

FAILURE STATES AND ANTI-PATTERNS

Shared control plane without recovery, guardrail drift, overprivileged support, consumer impact hidden, platform metric theater, no escape hatch, and platform outage treated as consumer fault.

RECOVERY AND RECONSIDERATION

Stabilize platform, protect consumers, preserve evidence, restore or fail over, provide alternate path, communicate, and reassess. Reopen on capability, consumer, policy, provider, or architecture change.

COMPLETION CRITERIA

Platform, capabilities, consumers, owners, policy, health, capacity, cost, evidence, support, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve platform state, consumer map, policies, incidents, changes, decisions, evidence, exceptions, and open gaps. Hand off to Platform Engineering, SRE, Security, Cloud, Support, Governance, and Operations.

RELATED CODEX SECTIONS

272 Platform Engineering; 277 Cloud; 280 Orchestration; 301 SRE; 304 Cloud Operations; 313-317 Observability; 323-330 Capacity, Availability, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, platform, cloud, operations, support, and owner-control context, and the integration roadmap. Exact platform, consumers, services, staffing, SLOs, cost, recovery, and acceptance authority remain unresolved.

311

SECTION 311 - SERVICE MANAGEMENT

CODEX/03_SECTIONS/311_SERVICE_MANAGEMENT.txt

SHA-256 898702c340332d1637c54c6d5c699882dee65eb06a81ca0d3e99b1929b65569c

4,984 BYTES / PRESERVED SOURCE IDENTITY

SECTION 311 — Service Management

STATUS: CONTENT-COMPLETE

PURPOSE

Govern the design, delivery, support, change, performance, continuity, and improvement of services from a user's and owner's outcome perspective.

REQUIRED OUTCOME

A service-management model with catalog, owner, consumer, level, request, incident, problem, change, knowledge, supplier, measure, continuity, and improvement.

WHY THIS MATTERS

Service work crosses technical and organizational boundaries. Without explicit service, consumer, support, and outcome contracts, teams measure components while users experience unresolved failure.

APPLICABILITY

Apply when a capability is provided as an ongoing service with consumers, support, expectations, incidents, requests, changes, or service commitments.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing service, consumer, support, commitment, or operational outcome exists and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, service levels, product, support, incident, change, problem, capacity, security, privacy, supplier, and continuity baselines.

REQUIRED INPUTS

Service, consumers, outcomes, catalog, level, support, requests, incidents, changes, suppliers, measures, costs, risks, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Service owners define outcomes, commitments, support, priority, change, incident, supplier, cost, and risk. Humans approve service levels, exceptions, communication, and retirement.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize service records, classify requests, identify trends, and draft knowledge or improvement. It must not make commitments, prioritize harm, close incidents, deny support, or change service without authority.

APPLICABLE EXISTING AGENT ROLES

Service Manager, Product Manager, Operations Engineer, Support Reviewer, Incident Responder, Problem Analyst, Researcher, Analyst, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define service and catalog; identify consumers; set levels and support; operate request, incident, problem, change, supplier, knowledge, continuity, measure, and improvement processes; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, consumer, outcome, level, request, incident, evidence, authority, and limitation. Require human review for priority, commitment, user impact, escalation, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link service, component, owner, dependency, contract, request, incident, problem, change, evidence, and level. Provide accessible support, status, communication, and recovery.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show service health, level, consumer impact, requests, incidents, changes, suppliers, cost, evidence freshness, owner, and authorized prioritize, communicate, recover, accept, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Service charter and catalog; consumer and support model; service levels; request, incident, problem, change, supplier, knowledge, continuity, decision, evidence, and improvement records.

EVIDENCE REQUIREMENTS

Identify service, consumer, outcome, level, period, source, method, result, actor, and limitation. Ticket closure or SLA percentage proves neither user resolution nor service quality.

VERIFICATION AND ACCEPTANCE TESTS

Verify catalog, ownership, support, request, incident, problem, change, communication, service level, supplier, accessibility, continuity, and user outcome.

FAILURE STATES AND ANTI-PATTERNS

Component metric as service, ticket closure theater, SLA gaming, no consumer voice, unowned supplier, support inaccessible, and incident without problem learning.

RECOVERY AND RECONSIDERATION

Protect users, communicate, restore service, correct records, escalate, and reassess commitment and residual impact. Reopen on user, service, supplier, level, or operational change.

COMPLETION CRITERIA

Service, consumers, outcomes, ownership, support, levels, records, measures, evidence, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner and consumer acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve catalog, contracts, tickets, incidents, changes, supplier records, decisions, evidence, exceptions, and open gaps. Hand off to Operations, Support, SRE, Product, Security, Governance, and Retirement.

RELATED CODEX SECTIONS

029 Operational Architecture; 301-310 SRE, Production, Runtime, Cloud, Network, Security, Data, AI, Application, and Platform Operations; 312 IT Operations; 318-330 Incident, Capacity, Availability, Performance, Reliability, Service Level, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, support, operations, and owner-control context, and the integration roadmap. Exact services, consumers, levels, staffing, suppliers, measures, and acceptance authority remain unresolved.

312

SECTION 312 - IT OPERATIONS

CODEX/03_SECTIONS/312_IT_OPERATIONS.txt

SHA-256 dea33b9ea7d8e3d731d3ba00fcd3eb0cec6cab34d6b5a0f3e7a4ccdb4272abe9

4,878 BYTES / PRESERVED SOURCE IDENTITY

SECTION 312 — IT Operations

STATUS: CONTENT-COMPLETE

PURPOSE

Operate organizational technology services, endpoints, accounts, infrastructure, support, access, change, security, and continuity for internal and external users.

REQUIRED OUTCOME

An IT-operations model with service catalog, users, assets, support, identity, devices, applications, network, incidents, changes, suppliers, continuity, and retirement.

WHY THIS MATTERS

IT operations affects daily work and organizational control. Unmanaged endpoints, weak support, stale access, supplier failure, and undocumented workarounds create security, productivity, and continuity risk.

APPLICABILITY

Apply when organizational IT services, devices, applications, accounts, support, or infrastructure are operated or changed.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no IT service, user, asset, access, support, or continuity behavior changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use service management, IAM, endpoint, network, security, privacy, asset, supplier, support, backup, and continuity baselines.

REQUIRED INPUTS

Services, users, assets, accounts, endpoints, applications, support, incidents, changes, suppliers, policies, evidence, continuity, and retirement.

HUMAN RESPONSIBILITIES AND AUTHORITY

IT owners approve service, access, device, support, change, incident, supplier, continuity, exception, and retirement. Humans protect user agency and accessible support.

AI RESPONSIBILITIES AND BOUNDARIES

AI may classify authorized tickets, summarize health, draft knowledge, and identify patterns. It must not access user content unnecessarily, revoke access, change devices, close sensitive tickets, or make employment or user-impact decisions without authority.

APPLICABLE EXISTING AGENT ROLES

IT Operations Engineer, Service Manager, Systems Administrator, Support Reviewer, Security Auditor, IAM Engineer, Network Engineer, Tester, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Operate services and assets; provision and deprovision; support users; manage access, devices, applications, changes, incidents, suppliers, backups, continuity, and retirement; measure outcomes and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show user, service, asset, ticket, access, action, evidence, privacy boundary, and limitation. Require human review for sensitive content, access, employment impact, device wipe, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use inventory, identity, endpoint management, least privilege, service catalog, support, knowledge, audit, privacy, backup, recovery, and accessible alternatives.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show IT service, asset, user impact, access, incidents, requests, suppliers, evidence freshness, owner, and authorized approve, support, restrict, recover, replace, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

IT operations charter; service and asset catalog; support and access; incident and change; supplier, continuity, recovery, decision, evidence, and retirement records.

EVIDENCE REQUIREMENTS

Identify service, user, asset, action, source, method, time, result, actor, privacy boundary, and limitation. Ticket status proves only workflow state.

VERIFICATION AND ACCEPTANCE TESTS

Verify service, support, access, device, application, network, security, privacy, backup, restore, continuity, supplier, and user recovery. Test inaccessible support and lost device.

FAILURE STATES AND ANTI-PATTERNS

Ticket closure as resolution, shared account, unmanaged device, inaccessible support, wipe without authority, supplier hidden, and workaround without record.

RECOVERY AND RECONSIDERATION

Support affected users, preserve evidence, restore or replace, revoke or recover access, communicate, and reassess privacy and continuity. Reopen on user, service, asset, supplier, or policy change.

COMPLETION CRITERIA

Services, users, assets, access, support, incidents, changes, suppliers, recovery, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve asset and service records, tickets, incidents, access, supplier, decisions, evidence, exceptions, and open gaps. Hand off to Service Management, Security, IAM, Support, Operations, Records, and Governance.

RELATED CODEX SECTIONS

200-202 IAM; 283 Systems Administration; 311 Service Management; 313-317 Observability; 328-330 Backup, Restore, and Failover; 396 Account Closure.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, user, asset, support, security, and operations context, and the integration roadmap. Exact users, services, assets, staffing, privacy, suppliers, recovery, and acceptance authority remain unresolved.

313

SECTION 313 - OBSERVABILITY ENGINEERING

CODEX/03_SECTIONS/313_OBSERVABILITY_ENGINEERING.txt

SHA-256 739f62aaaf99b94378a8f0cfa1fecfecee1c90f938669fdb2162c4e5cec76e3f

4,941 BYTES / PRESERVED SOURCE IDENTITY

SECTION 313 — Observability Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Design signals that allow humans and systems to understand internal and user-relevant behavior, state, failure, performance, security, and recovery from observable outputs.

REQUIRED OUTCOME

An observability model with questions, signals, sources, context, cardinality, privacy, retention, correlation, dashboards, alerts, ownership, cost, and known blind spots.

WHY THIS MATTERS

Collecting more telemetry does not guarantee understanding. Uncontextualized, noisy, private, expensive, or unowned signals can hide the actual failure while increasing risk.

APPLICABILITY

Apply when operators, owners, users, auditors, or automated systems need evidence about runtime, user, data, security, performance, or recovery behavior.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no material state, outcome, failure, control, or operational question needs new or changed observability and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, service objectives, privacy, security, logging, tracing, metrics, incident, cost, and data-retention baselines.

REQUIRED INPUTS

Operational questions, user journeys, services, signals, sources, context, schema, privacy, retention, cost, alert, owners, and blind spots.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define questions and acceptable collection; privacy and security owners approve data handling; operators use signals; humans decide response and tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate authorized telemetry, identify gaps, draft dashboards and alerts, and summarize behavior. It must not infer certainty from missing signals, expose private data, suppress evidence, or execute response without authority.

APPLICABLE EXISTING AGENT ROLES

Observability Engineer, SRE, Operations Engineer, Security Auditor, Privacy Reviewer, Data Engineer, Analyst, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define questions and outcomes; select signals; instrument; add context and correlation; protect and retain; build views and alerts; validate coverage and cost; document gaps; and review after incidents and change.

HUMAN-AI INTERACTION REQUIREMENTS

Show question, signal, source, context, privacy, time, evidence, inference, and limitation. Require human review for collection, alert, retention, user impact, and response.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use consistent identity and time, correlation, sampling, cardinality controls, redaction, access, resilient telemetry, independent evidence path, and separation of operational and sensitive content.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show observability coverage, questions, signal health, gaps, cost, privacy, retention, evidence freshness, owner, and authorized instrument, restrict, tune, retain, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Observability charter; question and signal catalog; instrumentation and schema; dashboards and alert; privacy and retention; gap, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify question, signal, source, context, timestamp, method, actor, result, coverage, privacy, and limitation. Signal presence proves only observed output and collection health.

VERIFICATION AND ACCEPTANCE TESTS

Verify coverage, context, identity, time, privacy, access, retention, cost, alert, correlation, missing signal, tamper, and recovery. Test whether a human can diagnose declared scenarios.

FAILURE STATES AND ANTI-PATTERNS

Telemetry volume as observability, no business context, high-cardinality cost, private data leakage, unowned dashboard, alert without action, and blind spot hidden.

RECOVERY AND RECONSIDERATION

Restore or substitute telemetry, preserve available evidence, reduce collection risk, notify owners, and reassess diagnosis. Reopen on service, question, signal, privacy, cost, or incident change.

COMPLETION CRITERIA

Questions, signals, context, privacy, owners, views, alerts, cost, retention, evidence, blind spots, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve instrumentation, schemas, dashboards, alerts, gaps, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Monitoring, Logging, Tracing, Metrics, SRE, Security, Data, and Operations.

RELATED CODEX SECTIONS

029 Operational Architecture; 301-312 Operations; 314-317 Monitoring, Logging, Tracing, and Metrics; 323-330 Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, operations, telemetry, privacy, security, and evidence context, and the integration roadmap. Exact questions, signals, cost, retention, owners, tools, and acceptance authority remain unresolved.

314

SECTION 314 - MONITORING

CODEX/03_SECTIONS/314_MONITORING.txt

SHA-256 390c0a6bfe6336e2e21db3668c12e6a780dc20930b66ada9936ddf2b1608ebff

4,463 BYTES / PRESERVED SOURCE IDENTITY

SECTION 314 — Monitoring

STATUS: CONTENT-COMPLETE

PURPOSE

Continuously observe defined states, thresholds, outcomes, and signals and route meaningful deviations to accountable humans or controlled automation.

REQUIRED OUTCOME

A monitoring model with targets, signals, thresholds, frequency, source, owner, alert action, escalation, privacy, retention, cost, and blind spots.

WHY THIS MATTERS

Monitoring fails when signals are noisy, thresholds arbitrary, actions unowned, or dashboards disconnected from users. No alert can mean healthy, blind, broken, or unmonitored.

APPLICABILITY

Apply when ongoing observation supports operations, security, safety, capacity, quality, compliance, or user outcomes.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no ongoing state or outcome needs observation and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use observability, metrics, logs, traces, service levels, incident, privacy, security, cost, and operations baselines.

REQUIRED INPUTS

Target, outcome, signal, source, threshold, frequency, context, owner, alert, escalation, retention, privacy, and response.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define what matters and response; operators tune and act; privacy and security owners approve collection; humans decide severity and action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize monitoring, identify anomalies, draft thresholds and alerts, and recommend investigation. It must not suppress, alter, attribute, or act on alerts beyond authority.

APPLICABLE EXISTING AGENT ROLES

Monitoring Engineer, SRE, Operations Engineer, Security Auditor, Analyst, Observer, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define targets; instrument; set thresholds; collect; validate; alert; triage; tune; protect and retain; test false positives and misses; review after change; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show target, signal, threshold, source, owner, action, evidence, privacy, and limitation. Require human approval for alert action, threshold, retention, and response.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use multi-signal health, business context, resilient collectors, time alignment, access, redaction, alert routing, and safe behavior when monitoring fails.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show target, signal health, threshold, alerts, owner, evidence freshness, cost, and authorized acknowledge, tune, mute with expiry, escalate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Monitoring charter; target and signal inventory; threshold and alert catalog; routing and escalation; test, incident, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify target, signal, source, threshold, period, method, result, owner, action, and limitation. Alert absence proves only the monitored signal and coverage.

VERIFICATION AND ACCEPTANCE TESTS

Verify signal, threshold, route, escalation, latency, privacy, retention, cost, false positive, false negative, collector failure, and action. Test detection of known scenarios.

FAILURE STATES AND ANTI-PATTERNS

Alert fatigue, mute without expiry, threshold by habit, no owner, collector single point, metric without action, and monitoring health confused with target health.

RECOVERY AND RECONSIDERATION

Restore collector, preserve evidence, use alternate signal, notify owners, and reassess blind period and impact. Reopen on target, signal, threshold, service, or incident change.

COMPLETION CRITERIA

Targets, signals, thresholds, owners, alerts, actions, evidence, privacy, retention, cost, blind spots, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve monitoring configs, alerts, incidents, gaps, decisions, evidence, exceptions, and open gaps. Hand off to Observability, Logging, Tracing, Metrics, SRE, Security, and Operations.

RELATED CODEX SECTIONS

313 Observability; 315-317 Logging, Tracing, and Metrics; 301-312 Operations; 318-330 Incident, Capacity, Availability, Performance, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, monitoring, observability, security, and operations context, and the integration roadmap. Exact targets, signals, thresholds, tools, staffing, privacy, and acceptance authority remain unresolved.

315

SECTION 315 - LOGGING

CODEX/03_SECTIONS/315_LOGGING.txt

SHA-256 1046b49c2ed6faa11823d8386dab7100e435b3a569581816a69864fa1c44c717

4,802 BYTES / PRESERVED SOURCE IDENTITY

SECTION 315 — Logging

STATUS: CONTENT-COMPLETE

PURPOSE

Record meaningful, structured, attributable, time-bound, privacy-aware events for operation, diagnosis, security, audit, compliance, and learning.

REQUIRED OUTCOME

A logging model with event taxonomy, fields, identity, time, correlation, severity, redaction, access, retention, integrity, transport, cost, and use.

WHY THIS MATTERS

Logs can be missing, misleading, excessive, sensitive, mutable, or impossible to correlate. More logging can increase privacy and cost while still failing to answer the operational question.

APPLICABILITY

Apply when event records support operation, security, audit, debugging, compliance, safety, user support, or forensic evidence.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no event evidence, operational question, audit, security, or user-support need changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use observability, privacy, security, records, incident, trace, metric, data, and operations baselines. Define log owner and retention authority.

REQUIRED INPUTS

Events, fields, identity, time, correlation, severity, source, data classification, redaction, destination, access, retention, integrity, and cost.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define required evidence and collection; privacy and security owners approve content and access; operators interpret logs; humans decide disclosure and retention.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize authorized logs, identify patterns, draft schemas and queries, and propose redaction. It must not expose sensitive content, delete evidence, infer certainty from silence, or alter retention without authority.

APPLICABLE EXISTING AGENT ROLES

Logging Engineer, Observability Engineer, Security Auditor, Privacy Reviewer, SRE, Data Engineer, Analyst, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define events and fields; instrument; normalize; timestamp; correlate; redact; transport and store; protect access and integrity; monitor collection; query; retain; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show log source, event, identity, time, data class, query, evidence, privacy, and limitation. Require human approval for sensitive query, disclosure, retention, and deletion.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use structured schema, time sync, correlation ID, resilient transport, encryption, access control, tamper evidence, redaction, sampling, retention tier, and independent evidence store.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show log coverage, source health, schema, privacy, retention, cost, incidents, evidence freshness, owner, and authorized query, restrict, preserve, redact, export, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Logging charter; event and schema catalog; redaction and access; pipeline and retention; integrity and query; incident, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, event, schema, identity, time basis, correlation, method, query, result, actor, and limitation. A log line proves only its recorded event and integrity context.

VERIFICATION AND ACCEPTANCE TESTS

Verify event coverage, fields, identity, time, correlation, redaction, access, retention, integrity, transport, query, collector loss, and evidence recovery.

FAILURE STATES AND ANTI-PATTERNS

Secrets in logs, personal data overcollection, no timestamp, uncorrelated events, mutable storage, log volume as observability, and log absence treated as no event.

RECOVERY AND RECONSIDERATION

Restore or substitute collection, preserve available records, rotate exposed secrets, restrict access, notify owners, and reassess blind period. Reopen on event, schema, privacy, incident, or retention change.

COMPLETION CRITERIA

Events, fields, source, identity, time, correlation, privacy, access, retention, integrity, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve schemas, configs, retention, queries, incidents, evidence, decisions, exceptions, and open gaps. Hand off to Observability, Security, Privacy, Tracing, Metrics, SRE, Records, and Operations.

RELATED CODEX SECTIONS

004 Evidence and Truth; 206 Security Operations; 213 Digital Forensics; 225 Records; 313-317 Observability, Monitoring, Tracing, and Metrics; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, evidence, privacy, security, observability, and operations context, and the integration roadmap. Exact events, fields, retention, tools, privacy, cost, and acceptance authority remain unresolved.

316

SECTION 316 - TRACING

CODEX/03_SECTIONS/316_TRACING.txt

SHA-256 84f18d3b7f73bc74e078ae74c1bd41a5e9b4124266239a70709872bd529f4f58

4,808 BYTES / PRESERVED SOURCE IDENTITY

SECTION 316 — Tracing

STATUS: CONTENT-COMPLETE

PURPOSE

Follow a request, transaction, workflow, job, or decision across components and boundaries with sufficient context to understand latency, failure, dependency, and user outcome.

REQUIRED OUTCOME

A tracing model with trace and span identity, propagation, boundaries, timing, attributes, sampling, privacy, storage, correlation, access, cost, and diagnostic use.

WHY THIS MATTERS

Without cross-boundary context, teams guess at causality. Traces can also leak payloads, create high cost, or imply complete causality when sampling and instrumentation are partial.

APPLICABILITY

Apply when distributed requests, jobs, events, workflows, tool calls, or decisions cross components and timing or causality matters.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no distributed path, dependency, diagnostic, or evidence claim changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use observability, logging, metrics, privacy, security, runtime, integration, incident, and data-retention baselines.

REQUIRED INPUTS

Trace, span, propagation, component, timing, attributes, sampling, payload policy, storage, access, retention, and diagnostic questions.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define diagnostic questions and context; privacy and security owners approve attributes and access; operators interpret traces; humans decide disclosure and action.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized traces, identify dependency patterns, draft queries, and summarize latency or failure. It must not expose payloads, infer causality beyond evidence, alter sampling, or act on a trace without authority.

APPLICABLE EXISTING AGENT ROLES

Tracing Engineer, Observability Engineer, SRE, Application Engineer, Network Engineer, Privacy Reviewer, Analyst, Tester, Verifier, and Incident Responder.

REQUIRED ACTIVITIES

Define path and boundaries; instrument; propagate identity; capture timing and context; sample and protect; correlate with logs and metrics; query; validate gaps; retain; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show trace, span, component, timing, attributes, sample, query, evidence, privacy, and limitation. Require human approval for payload access, sensitive query, sampling, retention, and response.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use context propagation, clock handling, stable IDs, bounded attributes, redaction, sampling, resilient collectors, trace-to-log and metric correlation, and access control.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show trace health, coverage, latency, spans, dependencies, sampling, privacy, cost, evidence freshness, owner, and authorized query, restrict, preserve, adjust, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Tracing charter; path and span model; propagation and sampling; privacy and access; correlation and query; incident, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify trace, span, component, clock, sample, attribute, method, time, result, actor, and limitation. A trace shows observed spans, not every unseen path or causal explanation.

VERIFICATION AND ACCEPTANCE TESTS

Verify propagation, identity, timing, sampling, privacy, access, correlation, collector loss, clock skew, dropped span, and diagnostic usefulness. Test known failure path.

FAILURE STATES AND ANTI-PATTERNS

Payload leakage, broken propagation, clock misread, sample treated as complete, high-cardinality cost, unowned traces, and causal claims from correlation alone.

RECOVERY AND RECONSIDERATION

Restore collection, preserve available traces, restrict payload access, adjust sampling, notify owners, and reassess diagnostic confidence. Reopen on path, component, privacy, sampling, or incident change.

COMPLETION CRITERIA

Paths, spans, identity, timing, sampling, privacy, access, correlation, evidence, cost, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve trace schemas, propagation, sampling, queries, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Observability, Logging, Metrics, Application, Network, Privacy, SRE, and Operations.

RELATED CODEX SECTIONS

034 Integration Architecture; 313 Observability; 314 Monitoring; 315 Logging; 317 Metrics; 301-310 Operations; 318-330 Incident, Performance, and Reliability.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, distributed systems, observability, privacy, and operations context, and the integration roadmap. Exact trace paths, context, sampling, payload, tools, cost, and acceptance authority remain unresolved.

317

SECTION 317 - METRICS ENGINEERING

CODEX/03_SECTIONS/317_METRICS_ENGINEERING.txt

SHA-256 139febc739409ce3a38745869a17c54d24c1530b9c10288263095099602f61ef

4,878 BYTES / PRESERVED SOURCE IDENTITY

SECTION 317 — Metrics Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Define, instrument, calculate, validate, govern, and retire measures that describe system, service, user, business, security, quality, cost, and operational outcomes.

REQUIRED OUTCOME

A metric model with definition, owner, unit, population, dimension, source, calculation, window, freshness, quality, threshold, action, lineage, and limitation.

WHY THIS MATTERS

Metrics create incentives and decisions. A precise-looking number can be semantically wrong, gamed, stale, biased, or disconnected from the outcome it claims to represent.

APPLICABILITY

Apply when a measure informs operation, quality, performance, service level, governance, product, security, safety, cost, or human decision.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no metric, decision, threshold, outcome, or evidence claim changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use data, analytics, observability, service-level, quality, user research, privacy, security, and governance baselines.

REQUIRED INPUTS

Question, outcome, definition, source, population, unit, dimensions, window, calculation, owner, threshold, action, freshness, and limitation.

HUMAN RESPONSIBILITIES AND AUTHORITY

Metric owners define meaning, use, threshold, and decision authority; domain owners validate semantics; humans approve consequential interpretation and resist metric gaming.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft metric definitions, inspect authorized data and telemetry, identify anomalies, and compare trends. It must not redefine meaning, infer causality, hide contrary measures, or make decisions from a metric without authority.

APPLICABLE EXISTING AGENT ROLES

Metrics Engineer, Data Analyst, Data Engineer, SRE, Product Analyst, Researcher, Privacy Reviewer, Verifier, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Define decision and outcome; establish metric contract; instrument source; calculate; validate population and lineage; set threshold and action; monitor quality and freshness; review incentives; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show definition, source, population, window, calculation, owner, threshold, evidence, uncertainty, and limitation. Require human review for semantic, causal, high-impact, and target-setting decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use metric catalog, semantic layer, lineage, versioning, quality checks, time and identity, dimensions, privacy, retention, alert, and reproducible calculation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show metric definition, source, freshness, quality, trend, threshold, owner, consumers, evidence, and authorized correct, restrict, annotate, retire, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Metric charter and catalog; source and lineage; calculation and quality; threshold and action; decision, evidence, limitation, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify metric, definition, source, population, window, version, method, time, result, owner, uncertainty, and limitation. A chart or number proves only its declared calculation and slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify meaning, grain, source, lineage, calculation, freshness, dimensions, privacy, threshold, alert, reproducibility, and relationship to outcome. Test missing and contradictory data.

FAILURE STATES AND ANTI-PATTERNS

Vanity metric, denominator drift, hidden filter, stale value, target gaming, causation from correlation, aggregate hiding harm, and metric as truth without source.

RECOVERY AND RECONSIDERATION

Annotate or withdraw affected metric, preserve versions, notify consumers, correct source or calculation, and reassess decisions. Reopen on definition, source, population, window, policy, or outcome change.

COMPLETION CRITERIA

Question, outcome, definition, source, population, calculation, owner, evidence, threshold, action, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve metric versions, definitions, lineage, data, dashboards, decisions, incidents, evidence, exceptions, and open gaps. Hand off to Analytics, Data, BI, SRE, Service Management, Product, and Governance.

RELATED CODEX SECTIONS

161-163 Analytics, BI, and Data Analysis; 239 Performance; 301 SRE; 313-316 Observability, Monitoring, Logging, and Tracing; 318-330 Service Levels, Capacity, Availability, Reliability, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data, analytics, observability, service, and decision context, and the integration roadmap. Exact outcomes, definitions, sources, owners, thresholds, privacy, and acceptance authority remain unresolved.

318

SECTION 318 - ALERTING

CODEX/03_SECTIONS/318_ALERTING.txt

SHA-256 f992ecff185b21835171b51f5c05ad10ecc25b5d5a525fc833a1f5fc0f801fc3

4,595 BYTES / PRESERVED SOURCE IDENTITY

SECTION 318 — Alerting

STATUS: CONTENT-COMPLETE

PURPOSE

Turn meaningful deviations from protected service, security, safety, capacity, quality, or data conditions into attributable, actionable, and proportionate notification.

REQUIRED OUTCOME

An alerting model with signal, condition, threshold, owner, severity, deduplication, routing, escalation, suppression, expiry, action, privacy, evidence, and feedback.

WHY THIS MATTERS

An alert is a request for attention, not proof of incident. Noisy, duplicated, stale, unowned, or un-actionable alerts create fatigue and make true danger easier to miss.

APPLICABILITY

Apply when a signal should prompt a human or controlled automation to investigate, decide, communicate, or act.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no deviation, response, or operational decision requires notification and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use monitoring, metrics, logs, traces, service levels, incident, security, privacy, staffing, and operations baselines.

REQUIRED INPUTS

Signal, condition, threshold, context, owner, severity, route, escalation, suppression, action, evidence, schedule, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define actionable conditions, severity, recipients, escalation, mute authority, and response. Humans decide whether an alert represents incident, risk, or expected variation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate authorized signals, group alerts, draft tuning, and recommend investigation. It must not suppress, close, escalate, or execute action without authority and must surface uncertainty and missing context.

APPLICABLE EXISTING AGENT ROLES

Alert Engineer, SRE, Operations Engineer, Security Operations Analyst, Analyst, Observer, Incident Responder, Tester, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Define outcome and action; configure; route; deduplicate; test; monitor delivery; tune; document runbook; review noise and misses; manage mute expiry; and retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show signal, condition, threshold, owner, severity, context, action, evidence, and limitation. Require human approval for mute, threshold, escalation, and consequential automation.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use multi-signal context, deduplication, rate limits, routing redundancy, escalation, expiry, audit, privacy, and safe behavior when alerting fails.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show alert, source, severity, owner, action, age, evidence freshness, mute, escalation, and authorized acknowledge, suppress with expiry, escalate, tune, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Alert charter; alert catalog; routing and escalation; runbook; noise and miss analysis; test, incident, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify signal, condition, threshold, source, context, time, route, actor, result, and limitation. Alert receipt proves delivery, not diagnosis or response.

VERIFICATION AND ACCEPTANCE TESTS

Verify trigger, threshold, context, deduplication, routing, escalation, mute expiry, privacy, delivery, false positive, missed event, and response. Exercise notification outage.

FAILURE STATES AND ANTI-PATTERNS

Alert fatigue, no owner, permanent mute, severity inflation, no runbook, alert as incident verdict, and automation that acts on an unverified signal.

RECOVERY AND RECONSIDERATION

Restore notification, use alternate channel, preserve signal and blind period, notify owners, and reassess missed impact. Reopen on target, threshold, owner, route, or incident change.

COMPLETION CRITERIA

Signals, conditions, thresholds, owners, actions, routes, evidence, privacy, mute, escalation, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve alert definitions, routes, mutes, incidents, evidence, decisions, exceptions, and open gaps. Hand off to Monitoring, Incident, Security, SRE, Operations, and Support.

RELATED CODEX SECTIONS

314 Monitoring; 317 Metrics; 319 Incident Management; 306 Security Operations; 301 SRE; 323-330 Performance, Reliability, Continuity, and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, monitoring, incident, operations, and owner-control context, and the integration roadmap. Exact signals, thresholds, staffing, routes, privacy, actions, and acceptance authority remain unresolved.

319

SECTION 319 - INCIDENT MANAGEMENT

CODEX/03_SECTIONS/319_INCIDENT_MANAGEMENT.txt

SHA-256 361929396a10ce61fd3cb0924e80ac8760e40d1a2ec5a50b0ef61579039fb7d3

4,862 BYTES / PRESERVED SOURCE IDENTITY

SECTION 319 — Incident Management

STATUS: CONTENT-COMPLETE

PURPOSE

Coordinate the lifecycle of operational incidents from detection and declaration through command, communication, stabilization, recovery, closure, and learning.

REQUIRED OUTCOME

An incident-management model with classification, commander, roles, timeline, evidence, impact, communications, actions, dependencies, recovery, closure, and review.

WHY THIS MATTERS

Incidents require coordinated authority under uncertainty. Unclear command, parallel unsafe action, missing timeline, or premature closure can increase harm and prevent recovery learning.

APPLICABILITY

Apply when an event disrupts or threatens service, safety, security, privacy, data, operations, users, property, or commitments.

NON-APPLICABILITY CONDITIONS

Record non-applicability only for events with no material impact and a routine path that preserves evidence, escalation, and user support.

PREREQUISITES AND DEPENDENCIES

Use alert, monitoring, service, security, continuity, communication, support, legal, privacy, records, and recovery baselines.

REQUIRED INPUTS

Alert or report, impact, scope, severity, commander, contacts, evidence, actions, dependencies, communications, recovery objective, and owner authority.

HUMAN RESPONSIBILITIES AND AUTHORITY

Incident commander coordinates and can stop unsafe action. Owners approve containment, communication, service reduction, legal escalation, recovery, and closure. Humans decide material severity and harm.

AI RESPONSIBILITIES AND BOUNDARIES

AI may summarize signals and timeline, draft hypotheses and status, and track authorized actions. It must not declare, contain, communicate, delete evidence, attribute, or close without authority.

APPLICABLE EXISTING AGENT ROLES

Incident Commander, Incident Responder, SRE, Security Operations Analyst, Forensic Analyst, Coordinator, Operations Engineer, Legal Advisor, Privacy Reviewer, Verifier, and Documentation Writer.

REQUIRED ACTIVITIES

Prepare; intake; declare; assign command; preserve evidence; assess impact; communicate; contain; stabilize; recover; monitor; close; review; and track corrective action.

HUMAN-AI INTERACTION REQUIREMENTS

Show incident, severity, impact, scope, commander, evidence, action, authority, communication, and uncertainty. Require human approval for containment, notification, recovery, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use case record, immutable timeline, evidence store, command channel, status page, action authority, privacy, audit, and recovery independent of the failed service where possible.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show incident state, commander, severity, scope, impact, actions, evidence freshness, dependencies, communication, residual risk, and authorized escalate, contain, communicate, recover, reopen, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Incident charter; timeline; impact and severity; command and communications; evidence; actions and recovery; post-incident review; decision, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify event, source, asset, user, time basis, action, actor, result, impact, confidence, and limitation. Confirmed, suspected, disproved, and unknown states remain distinct.

VERIFICATION AND ACCEPTANCE TESTS

Exercise declaration, command, paging, evidence, communication, containment, recovery, user support, legal and privacy escalation, closure, and post-incident follow-up.

FAILURE STATES AND ANTI-PATTERNS

No commander, severity by volume, parallel conflicting action, lost timeline, status theater, premature closure, blame, and communication that overclaims.

RECOVERY AND RECONSIDERATION

Stabilize, preserve evidence, recover, communicate, support users, and reassess impact and residual risk. Reopen on new evidence, scope, root cause, recovery failure, or harm.

COMPLETION CRITERIA

Command, severity, scope, evidence, actions, communication, recovery, closure, lessons, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve case, timeline, evidence, communications, actions, decisions, postmortem, exceptions, and open gaps. Hand off to Problem, Security, SRE, Support, Legal, Privacy, Records, and Governance.

RELATED CODEX SECTIONS

213-214 Digital Forensics and Incident Response; 306 Security Operations; 318 Alerting; 320 Problem Management; 326-330 Continuity, DR, Backup, Restore, and Failover; 402-404 Review.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, incident, evidence, continuity, and operations context, and the integration roadmap. Exact severity, command, notification, staffing, recovery, legal, privacy, and acceptance authority remain unresolved.

320

SECTION 320 - PROBLEM MANAGEMENT

CODEX/03_SECTIONS/320_PROBLEM_MANAGEMENT.txt

SHA-256 24482a2e9b7aa254776cc7d05b4f944b4354a2518ccb431be4622df41f306e7c

4,746 BYTES / PRESERVED SOURCE IDENTITY

SECTION 320 — Problem Management

STATUS: CONTENT-COMPLETE

PURPOSE

Identify and remove systemic causes of incidents, defects, risk, toil, and recurring service failure while preserving safe workarounds and evidence.

REQUIRED OUTCOME

A problem model with symptoms, incidents, cause hypotheses, evidence, known error, workaround, corrective action, owner, risk, verification, and prevention.

WHY THIS MATTERS

Closing incidents restores service but may leave the cause. Root-cause language can also overclaim when evidence is incomplete or blame is substituted for system learning.

APPLICABILITY

Apply when incidents, defects, vulnerabilities, recurring workarounds, or systemic risk warrant investigation beyond immediate restoration.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no recurring or systemic issue exists and an incident closure is sufficient for the bounded event.

PREREQUISITES AND DEPENDENCIES

Use incident, change, architecture, quality, vulnerability, reliability, data, support, and evidence baselines.

REQUIRED INPUTS

Incidents, symptoms, timeline, systems, changes, dependencies, evidence, hypotheses, workarounds, risks, owners, and corrective-action criteria.

HUMAN RESPONSIBILITIES AND AUTHORITY

Problem owners coordinate investigation and action; domain experts challenge hypotheses; owners approve risk, workaround, change, and closure. Humans avoid assigning causal certainty without evidence.

AI RESPONSIBILITIES AND BOUNDARIES

AI may correlate authorized incidents, draft hypotheses, identify patterns, and suggest experiments. It must distinguish correlation from cause, preserve uncertainty, and not change systems or close problems without authority.

APPLICABLE EXISTING AGENT ROLES

Problem Manager, Incident Responder, Analyst, Researcher, Architect, SRE, Tester, Verifier, Critic, Governance Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Link incidents; define symptom and impact; preserve evidence; investigate hypotheses; establish workaround or known error; design corrective action; change safely; verify; communicate; and capture prevention.

HUMAN-AI INTERACTION REQUIREMENTS

Show symptom, evidence, hypothesis, confidence, impact, workaround, action, owner, and limitation. Require human review for cause, risk, change, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Link incident, change, dependency, configuration, architecture, test, evidence, and corrective action. Support reproducible diagnosis, safe experiments, rollback, and knowledge retention.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show problem, linked incidents, cause status, evidence, workaround, owner, corrective action, risk, due date, and authorized assign, investigate, mitigate, remediate, reopen, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Problem record; incident links; hypothesis and evidence; known error and workaround; corrective action; verification, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify symptom, incident, system, change, source, method, time, hypothesis, result, confidence, and limitation. A plausible narrative is not causal proof.

VERIFICATION AND ACCEPTANCE TESTS

Verify evidence, reproducibility, workaround, corrective action, regression, user impact, change, monitoring, and recurrence. Independent challenge causal conclusion where material.

FAILURE STATES AND ANTI-PATTERNS

Root cause by assertion, blame, workaround as fix, problem backlog without owner, incident closure copied as solution, and corrective action without recurrence evidence.

RECOVERY AND RECONSIDERATION

Use safe workaround, preserve evidence, protect users, implement or roll back corrective change, and reassess recurrence. Reopen on repeat incident, new evidence, invalid cause, or changed system.

COMPLETION CRITERIA

Symptom, impact, evidence, hypothesis, workaround, corrective action, owner, verification, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve problem, incidents, evidence, workarounds, changes, decisions, corrective action, exceptions, and open gaps. Hand off to Incident, Change, Architecture, Quality, SRE, Security, and Operations.

RELATED CODEX SECTIONS

319 Incident Management; 289 Change Management; 228-230 Quality; 243-245 Reliability and Resilience; 386 Technical Debt; 402-404 Review and Lessons.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, incident, quality, evidence, and operations context, and the integration roadmap. Exact cause method, owners, recurrence, risk, change, and acceptance authority remain unresolved.

321

SECTION 321 - CAPACITY MANAGEMENT

CODEX/03_SECTIONS/321_CAPACITY_MANAGEMENT.txt

SHA-256 7ea1220b0bae6c4d84d1e7368340836fecbc08b868a15756aa2d436d56598d99

4,605 BYTES / PRESERVED SOURCE IDENTITY

SECTION 321 — Capacity Management

STATUS: CONTENT-COMPLETE

PURPOSE

Plan, measure, allocate, forecast, protect, and improve resources needed to meet service, user, safety, reliability, cost, and growth outcomes.

REQUIRED OUTCOME

A capacity model with resource, workload, demand, headroom, forecast, threshold, cost, scaling, dependency, owner, action, and failure plan.

WHY THIS MATTERS

Capacity failure often appears as latency, errors, queue growth, data loss, or unsafe overload before a simple utilization number turns red. Forecasts are assumptions and need validation.

APPLICABILITY

Apply when resources, demand, growth, concurrency, data, staff, provider quota, or budget can limit a service or outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no capacity, demand, resource, cost, or service objective changes and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use architecture, performance, scalability, SRE, cost, provider, staffing, data, operations, and service-level baselines.

REQUIRED INPUTS

Resources, demand, workload, growth, limits, headroom, service objective, cost, dependencies, forecast, scale, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners set service and cost tradeoffs, headroom, staffing, scaling, throttling, and investment. Humans accept forecast uncertainty and user impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized capacity and workload data, forecast bounded trends, and propose scale or throttle. It must expose assumptions and must not scale, shed users, or make commitments without authority.

APPLICABLE EXISTING AGENT ROLES

Capacity Engineer, SRE, Performance Engineer, Architect, FinOps Analyst, Data Analyst, Operations Engineer, Tester, Verifier, and Planner.

REQUIRED ACTIVITIES

Define service and resources; measure demand and utilization; establish headroom; forecast; test scale; plan procurement or automation; monitor; throttle or expand; review forecast error; and recover.

HUMAN-AI INTERACTION REQUIREMENTS

Show workload, resource, forecast, assumption, threshold, cost, service impact, evidence, and limitation. Require human approval for scale, throttle, budget, and commitment decisions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use quotas, autoscaling guardrails, queues, backpressure, partitioning, caching, capacity signals, resource isolation, cost controls, and graceful degradation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show demand, resource, headroom, forecast, cost, SLO, incidents, evidence freshness, owner, and authorized scale, throttle, defer, budget, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Capacity charter; resource and workload model; forecast and headroom; scaling and budget plan; test, incident, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify resource, workload, period, forecast, source, method, threshold, result, cost, uncertainty, and limitation. A utilization average or forecast proves only its conditions.

VERIFICATION AND ACCEPTANCE TESTS

Verify demand model, measurement, headroom, threshold, scale, throttle, cost, dependency, queue, overload, forecast error, and recovery. Test beyond expected scale safely.

FAILURE STATES AND ANTI-PATTERNS

Utilization-only, no headroom, linear forecast, autoscaling cost surprise, capacity hidden in vendor quota, throttle without fairness, and commitment without evidence.

RECOVERY AND RECONSIDERATION

Protect critical users, throttle or scale, preserve evidence, restore service, communicate, and reassess architecture and forecast. Reopen on workload, growth, provider, cost, or service change.

COMPLETION CRITERIA

Resources, demand, workload, forecast, headroom, cost, service, owners, evidence, actions, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve capacity history, forecasts, incidents, scaling, budgets, decisions, evidence, exceptions, and open gaps. Hand off to Performance, SRE, Cloud, Platform, Finance, and Operations.

RELATED CODEX SECTIONS

239-242 Performance, Load, Stress, and Scalability; 301 SRE; 317 Metrics; 322-324 Availability, Performance, and Reliability; 304 Cloud Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, capacity, performance, cost, SRE, and operations context, and the integration roadmap. Exact demand, resources, forecasts, budgets, service objectives, staffing, and acceptance authority remain unresolved.

322

SECTION 322 - AVAILABILITY MANAGEMENT

CODEX/03_SECTIONS/322_AVAILABILITY_MANAGEMENT.txt

SHA-256 cf9a118deb70c6b48770b41df7034c2d197f11b9cda695d7fdb7c7c56862a9e2

4,658 BYTES / PRESERVED SOURCE IDENTITY

SECTION 322 — Availability Management

STATUS: CONTENT-COMPLETE

PURPOSE

Ensure services and capabilities are available to authorized users and dependent systems when needed, with declared windows, degradation, maintenance, and recovery.

REQUIRED OUTCOME

An availability model with service, critical journey, window, objective, dependency, redundancy, maintenance, monitoring, incident, recovery, and evidence.

WHY THIS MATTERS

Availability is not merely process uptime. A service can respond while unusable, inaccessible, incorrect, unsafe, or unavailable to a critical population.

APPLICABILITY

Apply when service access, continuity, user journey, dependency, maintenance, or recovery is material.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no availability, access, dependency, maintenance, or recovery outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use service levels, SRE, architecture, capacity, performance, resilience, continuity, support, monitoring, incident, and recovery baselines.

REQUIRED INPUTS

Service, users, critical journeys, window, objective, dependency, redundancy, maintenance, signals, incidents, support, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define availability and user outcomes, maintenance, redundancy, incident, service reduction, communication, and recovery. Humans accept availability and cost tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized health and outage data, identify dependencies, and recommend bounded action. It must not declare available, suppress incident, fail over, or communicate status without authority.

APPLICABLE EXISTING AGENT ROLES

Availability Engineer, SRE, Operations Engineer, Architect, Capacity Engineer, Incident Responder, Tester, Verifier, and Service Manager.

REQUIRED ACTIVITIES

Define outcomes and windows; map dependencies; design redundancy and maintenance; monitor; manage incident; fail over or reduce service; recover; measure; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, journey, window, dependency, signal, user impact, evidence, authority, and limitation. Require human approval for failover, maintenance, status, and service reduction.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use redundancy, health and business checks, graceful degradation, maintenance mode, failover, rollback, queueing, backup, and independent control.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show service availability, journey, dependency, maintenance, incident, evidence freshness, owner, and authorized pause, fail over, reduce, restore, communicate, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Availability charter; service and journey; dependency and redundancy; maintenance and monitoring; incident, failover, recovery, decision, evidence, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, journey, window, source, metric, period, method, result, user impact, and limitation. Uptime or ping proves only reachability and selected interval.

VERIFICATION AND ACCEPTANCE TESTS

Verify availability objective, critical journey, dependency, health, maintenance, redundancy, failover, communication, recovery, and user outcome. Test partial and regional outage.

FAILURE STATES AND ANTI-PATTERNS

Uptime-only, critical journey omitted, dependency hidden, failover untested, maintenance surprise, status page inaccurate, and availability traded for unsafe operation.

RECOVERY AND RECONSIDERATION

Stabilize, fail over or reduce service, preserve evidence, communicate, restore, and reassess user and commitment impact. Reopen on service, dependency, window, architecture, or recovery change.

COMPLETION CRITERIA

Service, journey, window, objective, dependency, redundancy, evidence, incident, recovery, owner, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve availability history, maintenance, incidents, failovers, decisions, evidence, exceptions, and open gaps. Hand off to SRE, Capacity, Performance, Continuity, Disaster Recovery, Support, and Operations.

RELATED CODEX SECTIONS

301 SRE; 321 Capacity; 323-324 Performance and Reliability; 325 Service Levels; 326-330 Continuity, DR, Backup, Restore, and Failover.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, availability, service, SRE, resilience, and operations context, and the integration roadmap. Exact services, journeys, windows, objectives, dependencies, recovery, and acceptance authority remain unresolved.

323

SECTION 323 - PERFORMANCE MANAGEMENT

CODEX/03_SECTIONS/323_PERFORMANCE_MANAGEMENT.txt

SHA-256 4e139eb41371823582330c91e30ab2286b86a7a5e17c9bf3802ddcbb01fae9c3

4,408 BYTES / PRESERVED SOURCE IDENTITY

SECTION 323 — Performance Management

STATUS: CONTENT-COMPLETE

PURPOSE

Manage response, throughput, resource, latency, cost, and user-experience performance over time through objectives, monitoring, capacity, change, and improvement.

REQUIRED OUTCOME

A performance-management model with service and user outcomes, indicators, objectives, workload, baselines, budgets, ownership, action, and review.

WHY THIS MATTERS

Performance changes with workload, dependency, data, configuration, and time. A one-time test or average can miss tail behavior and can encourage optimization that harms correctness or cost.

APPLICABILITY

Apply when performance affects users, operators, service levels, cost, capacity, safety, or business outcome.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no performance, workload, cost, or user outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use performance tests, metrics, SRE, capacity, architecture, cost, service levels, and operations baselines.

REQUIRED INPUTS

Outcome, workload, indicators, objective, baseline, latency, throughput, resource, cost, dependency, incidents, and action.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners define meaningful performance and tradeoffs; engineers manage causes; humans decide scale, degrade, cost, and acceptance.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized performance data, identify bottlenecks, and propose bounded tuning. It must expose context and not change production or optimize a proxy without authority.

APPLICABLE EXISTING AGENT ROLES

Performance Engineer, SRE, Capacity Engineer, Architect, Analyst, Operations Engineer, Tester, Verifier, Optimizer, and Documentation Writer.

REQUIRED ACTIVITIES

Define outcomes and indicators; baseline; monitor; analyze workload and dependencies; tune; capacity-plan; test; communicate; review regressions; and recover.

HUMAN-AI INTERACTION REQUIREMENTS

Show metric, workload, percentile, environment, dependency, cost, action, evidence, and limitation. Require human approval for tuning, scale, service reduction, and commitment.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use business and technical indicators, tail latency, resource and cost signals, backpressure, caching, queueing, degradation, capacity, and rollback.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show performance, workload, SLO, resource, cost, bottleneck, trend, evidence freshness, owner, and authorized tune, scale, throttle, rollback, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Performance charter; indicator and baseline; workload and capacity; tuning and change; incident, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, workload, metric, percentile, environment, period, source, method, result, cost, and limitation. Average latency or throughput proves only its measured slice.

VERIFICATION AND ACCEPTANCE TESTS

Verify indicators, workload, latency distribution, resource, cost, dependency, user outcome, regression, degradation, recovery, and reproducibility.

FAILURE STATES AND ANTI-PATTERNS

Average-only, benchmark theater, optimization without user outcome, cost hidden, warm cache, tail latency ignored, and performance target treated as quality.

RECOVERY AND RECONSIDERATION

Reduce workload, scale, roll back tuning, preserve evidence, communicate, and reassess architecture and objectives. Reopen on workload, dependency, version, cost, or user change.

COMPLETION CRITERIA

Outcomes, workload, indicators, objectives, resources, cost, evidence, owners, actions, limits, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve baselines, workload, tuning, incidents, metrics, decisions, evidence, exceptions, and open gaps. Hand off to Capacity, SRE, Testing, Application, Platform, and Operations.

RELATED CODEX SECTIONS

239-242 Performance and Scale Testing; 317 Metrics; 321 Capacity; 322 Availability; 324 Reliability; 325 Service Levels; 301-310 Operations.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, performance, capacity, SRE, service, and operations context, and the integration roadmap. Exact outcomes, workloads, indicators, objectives, costs, and acceptance authority remain unresolved.

324

SECTION 324 - RELIABILITY MANAGEMENT

CODEX/03_SECTIONS/324_RELIABILITY_MANAGEMENT.txt

SHA-256 1b59d7210068883f8e1ab6055eb27858e56ecd3b96666c54a4b5172d81fc63b8

4,670 BYTES / PRESERVED SOURCE IDENTITY

SECTION 324 — Reliability Management

STATUS: CONTENT-COMPLETE

PURPOSE

Manage reliability as a sustained outcome through failure prevention, detection, response, recovery, maintenance, change, capacity, and learning.

REQUIRED OUTCOME

A reliability model with user and service outcomes, failure modes, indicators, objectives, error budget, dependencies, maintenance, incident, recovery, and improvement.

WHY THIS MATTERS

Reliability degrades through change, dependency, workload, aging, operator burden, and hidden failure. A clean period does not prove future reliability or that recovery works.

APPLICABILITY

Apply when service or system reliability, durability, correctness, continuity, or recovery is material.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no reliability, failure, maintenance, or recovery outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use SRE, reliability testing, resilience, availability, performance, capacity, incident, problem, maintenance, and recovery baselines.

REQUIRED INPUTS

Service, user outcome, failure modes, indicators, objectives, budget, dependencies, maintenance, incidents, tests, recovery, and owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners set reliability objectives, budget, change, maintenance, staffing, incident, recovery, and risk. Humans decide tradeoffs and protect sustainable operation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze authorized failures and indicators, identify patterns, and draft improvement. It must not suppress failure, alter service, close incident, or accept reliability without authority.

APPLICABLE EXISTING AGENT ROLES

Reliability Engineer, SRE, Architect, Problem Manager, Incident Responder, Capacity Engineer, Tester, Verifier, Analyst, and Critic.

REQUIRED ACTIVITIES

Define outcomes and modes; identify failure; set objectives; monitor; test; manage error budget; maintain; respond; recover; reduce toil; review incidents; and improve architecture and process.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, outcome, failure, indicator, objective, budget, evidence, action, owner, and limitation. Require human decision for change, risk, incident, recovery, and budget.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use redundancy, graceful degradation, health, recovery, rollback, dependency isolation, capacity, maintenance, and independent control and evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show reliability, failures, SLO, budget, dependencies, incidents, maintenance, evidence freshness, owner, and authorized pause, rollback, scale, recover, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Reliability charter; failure and dependency map; objectives and budget; test and maintenance; incident, recovery, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, outcome, failure, period, indicator, source, method, result, owner, uncertainty, and limitation. Reliability estimate requires workload, duration, and exposure context.

VERIFICATION AND ACCEPTANCE TESTS

Verify objective, failure mode, indicators, tests, maintenance, dependency, error budget, recovery, incident learning, and operator workload. Exercise failure and degraded mode.

FAILURE STATES AND ANTI-PATTERNS

Uptime-only, reliability as blame, no failure budget, untested recovery, dependency hidden, toil ignored, and green period treated as durable reliability.

RECOVERY AND RECONSIDERATION

Stabilize, reduce change, fail over or restore, preserve evidence, communicate, and reassess objectives and architecture. Reopen on failure, workload, dependency, service, or recovery change.

COMPLETION CRITERIA

Outcomes, failure modes, objectives, budget, owners, dependencies, evidence, recovery, maintenance, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve reliability history, incidents, tests, maintenance, budgets, decisions, evidence, exceptions, and open gaps. Hand off to SRE, Availability, Performance, Capacity, Continuity, and Operations.

RELATED CODEX SECTIONS

243-245 Reliability, Resilience, and Chaos Testing; 301 SRE; 321-323 Capacity, Availability, and Performance; 325-330 Service Levels, Continuity, DR, Backup, Restore, and Failover.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, reliability, SRE, resilience, and operations context, and the integration roadmap. Exact outcomes, failure model, indicators, budgets, staffing, recovery, and acceptance authority remain unresolved.

325

SECTION 325 - SERVICE LEVEL MANAGEMENT

CODEX/03_SECTIONS/325_SERVICE_LEVEL_MANAGEMENT.txt

SHA-256 decdddfbc8ca0190c8cf75b9fa9ec246139f8601964ee294e0d5e6f9582165ba

4,644 BYTES / PRESERVED SOURCE IDENTITY

SECTION 325 — Service Level Management

STATUS: CONTENT-COMPLETE

PURPOSE

Define, negotiate, measure, review, communicate, and improve service expectations in terms users and owners can understand and evidence.

REQUIRED OUTCOME

A service-level model with service, consumer, outcome, indicator, objective, window, exclusions, measurement, commitment, escalation, remedy, and review.

WHY THIS MATTERS

Service levels can create clarity or encourage gaming. An availability percentage without user journey, error, latency, support, accessibility, or exclusion context can misrepresent service quality.

APPLICABILITY

Apply when a service commitment, internal objective, contract, support expectation, or measurable outcome is declared or changed.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no service expectation, consumer, metric, commitment, or operational outcome changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use service management, SRE, availability, performance, reliability, capacity, support, legal, contract, and metrics baselines.

REQUIRED INPUTS

Service, consumers, outcomes, indicators, objectives, windows, exclusions, source, measurement, commitment, escalation, remedy, and owner.

HUMAN RESPONSIBILITIES AND AUTHORITY

Service owners and consumers agree meaning, commitment, exclusions, escalation, remedies, and acceptance. Humans approve contractual language and risk tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may draft service-level definitions, analyze evidence, and identify gaps or trend. It must not make commitments, redefine exclusions, claim compliance, or communicate contract position without human authority.

APPLICABLE EXISTING AGENT ROLES

Service Manager, SRE, Metrics Engineer, Product Manager, Legal Advisor, Analyst, Verifier, Support Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Define service and outcome; identify consumers; set indicators and windows; agree objective; measure; report; review exclusions and remedies; manage breach; and improve or retire.

HUMAN-AI INTERACTION REQUIREMENTS

Show service, consumer, outcome, metric, window, objective, exclusion, evidence, and limitation. Require human approval for commitment, breach, remedy, and public communication.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use semantic metric contracts, source lineage, user and technical indicators, measurement integrity, audit, status, support, and clear exclusion handling.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show objective, period, result, evidence freshness, exclusions, consumer impact, owner, breach, remedy, and authorized correct, communicate, escalate, accept, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Service-level charter; consumer and outcome map; indicator and objective; measurement and reporting; breach and remedy; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify service, consumer, metric, window, source, exclusion, method, result, impact, and limitation. A percentage without denominator and exclusions proves little.

VERIFICATION AND ACCEPTANCE TESTS

Verify definition, source, window, calculation, exclusions, user outcome, reporting, breach, communication, remedy, and reproducibility. Test metric outage and disputed result.

FAILURE STATES AND ANTI-PATTERNS

SLA gaming, hidden exclusions, infrastructure proxy, commitment without capacity, breach not communicated, and metric treated as user truth.

RECOVERY AND RECONSIDERATION

Correct measurement, communicate, remediate, apply agreed remedy, preserve evidence, and reassess commitment. Reopen on service, consumer, metric, contract, or objective change.

COMPLETION CRITERIA

Service, consumer, outcome, metric, objective, window, exclusions, commitment, evidence, remedies, owners, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner and consumer acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve level versions, measurements, reports, breaches, communications, decisions, evidence, exceptions, and open gaps. Hand off to Service Management, SRE, Metrics, Legal, Support, and Operations.

RELATED CODEX SECTIONS

311 Service Management; 317 Metrics; 321-324 Capacity, Availability, Performance, and Reliability; 326-330 Continuity and Recovery.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, service, metrics, SRE, legal, and operations context, and the integration roadmap. Exact consumers, outcomes, objectives, exclusions, contracts, remedies, and acceptance authority remain unresolved.

326

SECTION 326 - BUSINESS CONTINUITY

CODEX/03_SECTIONS/326_BUSINESS_CONTINUITY.txt

SHA-256 c80f29fec3e4943d11bd84866e31e7307b44ceef8643a4bec34dfe22354ab505

5,070 BYTES / PRESERVED SOURCE IDENTITY

SECTION 326 — Business Continuity

STATUS: CONTENT-COMPLETE

PURPOSE

Preserve essential business, human, operational, legal, financial, and service outcomes through disruption, degradation, absence, dependency loss, and recovery.

REQUIRED OUTCOME

A continuity model with essential functions, priorities, people, facilities, dependencies, minimum service, alternate process, communication, recovery objective, exercise, and decision authority.

WHY THIS MATTERS

Continuity is about what must continue and how people work, not only how systems restart. A plan can fail through staffing, suppliers, communications, inaccessible alternatives, or untested assumptions.

APPLICABILITY

Apply when disruption could materially affect essential operations, obligations, people, customers, safety, finance, data, or public trust.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no essential outcome or dependency can be materially disrupted within scope and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use business impact, operational, disaster recovery, service, people, supplier, facility, data, communication, safety, and legal baselines.

REQUIRED INPUTS

Essential functions, priorities, users, staff, facilities, suppliers, systems, data, minimum service, dependencies, communication, recovery, and exercise evidence.

HUMAN RESPONSIBILITIES AND AUTHORITY

Continuity owners define essential outcomes, priorities, authority, staffing, alternate work, communication, and acceptance. Humans make life, safety, service, and resource tradeoffs.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map dependencies, summarize plans, draft exercises, and identify gaps. It must not declare continuity, reprioritize essential outcomes, communicate emergency instruction, or act on behalf of authorities without human control.

APPLICABLE EXISTING AGENT ROLES

Continuity Planner, Business Analyst, Operations Engineer, Coordinator, Risk Analyst, Researcher, Tester, Verifier, Safety Reviewer, and Documentation Writer.

REQUIRED ACTIVITIES

Identify essential functions and impact; set priority and minimum service; map people and dependencies; design alternate process and facility; communicate; exercise; recover; review; and update.

HUMAN-AI INTERACTION REQUIREMENTS

Show function, priority, dependency, people, minimum service, authority, evidence, and limitation. Require human approval for emergency action, resource allocation, public communication, and return to normal.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Support alternate channels, manual operation, resilient communications, data access, staffing, supplier substitution, dependency independence, accessible procedures, and recovery evidence.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show continuity mode, essential functions, priorities, resources, staffing, dependencies, communications, evidence freshness, and authorized activate, reprioritize, communicate, recover, and close controls.

REQUIRED OUTPUTS AND ARTIFACTS

Continuity charter; impact and essential-function map; alternate process and resource; communication and exercise; recovery, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify function, disruption, priority, people, dependency, method, time, result, impact, recovery, and limitation. A plan or tabletop proves only the exercised assumptions.

VERIFICATION AND ACCEPTANCE TESTS

Exercise staffing, facility, supplier, communication, manual work, data, accessibility, prioritization, alternate service, recovery, and return to normal. Record actual gaps.

FAILURE STATES AND ANTI-PATTERNS

System-only plan, no priority, unrealistic staffing, inaccessible procedures, supplier assumption, communication failure, tabletop theater, and continuity that shifts harm without decision.

RECOVERY AND RECONSIDERATION

Activate alternate work, protect people, communicate, preserve evidence, recover essential service, and reassess priority and residual impact. Reopen on function, people, dependency, facility, or threat change.

COMPLETION CRITERIA

Essential functions, priorities, people, dependencies, alternatives, authority, exercises, evidence, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve plans, contacts, exercises, decisions, communications, gaps, recovery, exceptions, and open actions. Hand off to Disaster Recovery, Operations, HR, Facilities, Suppliers, Legal, and Governance.

RELATED CODEX SECTIONS

005 Continuity and Recovery; 029 Operational Architecture; 018 Risk; 327 Disaster Recovery; 328-330 Backup, Restore, and Failover; 331 Operational Readiness.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, business impact, continuity, operations, people, and recovery context, and the integration roadmap. Exact functions, priorities, staff, suppliers, facilities, objectives, and acceptance authority remain unresolved.

327

SECTION 327 - DISASTER RECOVERY

CODEX/03_SECTIONS/327_DISASTER_RECOVERY.txt

SHA-256 84a989f775dd9219141e01ab4f277c8985b7f2bba63a0d03958f86244f02da5d

5,034 BYTES / PRESERVED SOURCE IDENTITY

SECTION 327 — Disaster Recovery

STATUS: CONTENT-COMPLETE

PURPOSE

Restore essential systems, data, services, and control after a disaster or major loss of site, provider, infrastructure, people, or trust.

REQUIRED OUTCOME

A disaster-recovery model with scenarios, priorities, recovery time and data objectives, dependencies, runbooks, authority, alternate environment, evidence, testing, and return to service.

WHY THIS MATTERS

Disaster recovery is an operational claim that must survive real dependency and authority loss. A backup or plan without a tested restore, staffing, access, or reconciliation path is not recovery capability.

APPLICABILITY

Apply when loss or unavailability of a system, facility, region, provider, data, identity, or operational capability could materially affect essential outcomes.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no disaster scenario, essential service, data, dependency, or recovery claim is material and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use continuity, backup, restore, failover, service, data, identity, security, facilities, supplier, communication, and operational readiness baselines.

REQUIRED INPUTS

Disaster scenarios, services, priorities, RTO, RPO, dependencies, alternate site, data, access, runbooks, staff, communication, exercises, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Recovery owners and commanders approve declaration, priority, failover, data loss, communication, resource, and return-to-service. Humans accept residual impact and legal duties.

AI RESPONSIBILITIES AND BOUNDARIES

AI may map dependencies, draft recovery steps, analyze exercise evidence, and identify gaps. It must not declare disaster, fail over, restore, overwrite data, or communicate emergency status without authority.

APPLICABLE EXISTING AGENT ROLES

Disaster Recovery Planner, Incident Commander, Operations Engineer, SRE, Data Engineer, Security Auditor, Continuity Planner, Tester, Verifier, and Coordinator.

REQUIRED ACTIVITIES

Define scenarios and priorities; establish RTO/RPO; map dependencies; prepare alternate environment and access; back up; exercise; declare; recover; reconcile; communicate; return; and improve.

HUMAN-AI INTERACTION REQUIREMENTS

Show scenario, priority, service, data, authority, RTO/RPO, action, evidence, and limitation. Require human approval for declaration, failover, restore, data loss, communication, and closure.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use alternate control path, independent credentials, replicated data, immutable backups, infrastructure rebuild, dependency substitution, reconciliation, staged return, and evidence preservation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show disaster mode, service priority, recovery phase, data point, dependencies, owner, evidence freshness, and authorized declare, prioritize, fail over, restore, communicate, and return controls.

REQUIRED OUTPUTS AND ARTIFACTS

DR charter; scenario and priority map; RTO/RPO; recovery architecture and runbooks; exercise results; data reconciliation; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify scenario, service, version, backup, restore, data point, time, authority, method, result, impact, and limitation. A tabletop or backup list proves only planning or inventory.

VERIFICATION AND ACCEPTANCE TESTS

Exercise declaration, access, alternate environment, backup, restore, dependencies, data loss, reconciliation, communication, staffing, security, and return. Record actual RTO/RPO and gaps.

FAILURE STATES AND ANTI-PATTERNS

Recovery dependent on failed system, backup not restored, credentials unavailable, data reconciliation omitted, priority unclear, tabletop only, and recovery success declared before user outcome.

RECOVERY AND RECONSIDERATION

Protect people and essential service, preserve evidence, recover by priority, reconcile, communicate, monitor, and reassess residual impact. Reopen on scenario, dependency, data, provider, or objective change.

COMPLETION CRITERIA

Scenarios, priorities, dependencies, RTO/RPO, access, data, runbooks, evidence, exercises, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve DR plans, backups, exercises, access, recoveries, decisions, communications, evidence, exceptions, and open gaps. Hand off to Continuity, Backup, Restore, Failover, Security, Data, and Operations.

RELATED CODEX SECTIONS

326 Business Continuity; 328-330 Backup, Restore, and Failover; 331 Operational Readiness; 005 Continuity and Recovery; 301 SRE; 319 Incident.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, disaster, continuity, data, security, and operations context, and the integration roadmap. Exact scenarios, services, RTO/RPO, alternate sites, staffing, dependencies, and acceptance authority remain unresolved.

328

SECTION 328 - BACKUP ENGINEERING

CODEX/03_SECTIONS/328_BACKUP_ENGINEERING.txt

SHA-256 4e9791acf73ad244a10d2f09b763075d3a613b81bca90a75d688b7194bf38eb0

4,827 BYTES / PRESERVED SOURCE IDENTITY

SECTION 328 — Backup Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Create protected, attributable, complete, recoverable, and lifecycle-governed copies of data, configuration, artifacts, and state needed for continuity and recovery.

REQUIRED OUTCOME

A backup model with scope, source, frequency, consistency, retention, encryption, isolation, immutability, ownership, integrity, monitoring, restore test, and destruction.

WHY THIS MATTERS

A completed backup job may capture the wrong data, a corrupted state, an inaccessible key, or a copy that cannot be restored. Backup can also preserve unwanted sensitive data indefinitely.

APPLICABILITY

Apply when data or state must be recovered after deletion, corruption, outage, disaster, ransomware, error, or migration.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no recoverable state, continuity obligation, data, configuration, or evidence requires backup and an approved exact baseline supports it.

PREREQUISITES AND DEPENDENCIES

Use data, storage, database, security, key, records, retention, continuity, DR, restore, and operations baselines.

REQUIRED INPUTS

Data and state, owner, priority, RPO, frequency, consistency, destination, encryption, key, retention, immutability, monitoring, restore, and deletion.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve scope, RPO, retention, encryption, access, restore, legal hold, cost, and destruction. Humans decide whether backup is sufficient evidence for recovery.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized backup metadata, jobs, and integrity results; draft plans and tests; and identify gaps. It must not read sensitive content, change scope, delete copies, restore, or expose keys without authority.

APPLICABLE EXISTING AGENT ROLES

Backup Engineer, Data Engineer, Storage Administrator, Database Administrator, Security Auditor, Continuity Planner, Tester, Verifier, and Operations Engineer.

REQUIRED ACTIVITIES

Classify recoverable state; define RPO and frequency; capture; encrypt and isolate; verify integrity; monitor; retain; test restore; manage key; delete; and report gaps.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, scope, RPO, destination, key reference, integrity, retention, evidence, and limitation without exposing content or keys. Require human approval for restore, deletion, scope, and retention.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use immutable or air-gapped copies where warranted, independent credentials, encryption, consistency, versioning, integrity checks, geographic separation, retention, and restore path.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show backup scope, age, RPO, integrity, destination, retention, key state, evidence freshness, owner, and authorized run, preserve, restore, restrict, delete, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Backup charter; scope and RPO; schedule and destination; protection and integrity; restore test; retention and deletion; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, backup version, scope, time, destination, integrity, key reference, method, result, owner, and limitation. Job success proves neither completeness nor restore capability.

VERIFICATION AND ACCEPTANCE TESTS

Verify scope, consistency, encryption, access, immutability, integrity, retention, key, restore, RPO, deletion, and incident behavior. Restore representative and critical data.

FAILURE STATES AND ANTI-PATTERNS

Backup not restored, same failure domain, key unavailable, retention unbounded, incomplete scope, backup admin shared, and backup dashboard treated as recoverability.

RECOVERY AND RECONSIDERATION

Protect copies, preserve evidence, restore or replicate, rotate keys, reconcile, notify owners, and reassess RPO and data impact. Reopen on data, source, key, destination, policy, or recovery change.

COMPLETION CRITERIA

Scope, source, RPO, frequency, protection, integrity, retention, restore, owners, evidence, deletion, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve backup inventory, jobs, versions, integrity, restore tests, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Restore, DR, Storage, Data, Security, Records, and Operations.

RELATED CODEX SECTIONS

225 Records; 286 Storage; 307 Data Operations; 326-330 Continuity, DR, Backup, Restore, and Failover; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, data, storage, security, continuity, and operations context, and the integration roadmap. Exact data, RPO, frequency, destinations, keys, retention, restore, and acceptance authority remain unresolved.

329

SECTION 329 - RESTORE ENGINEERING

CODEX/03_SECTIONS/329_RESTORE_ENGINEERING.txt

SHA-256 952ddc5b602cd96315dcdf0ed39180778c7afc66eba45e34a67c6bc5b64a35d8

4,804 BYTES / PRESERVED SOURCE IDENTITY

SECTION 329 — Restore Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Restore data, configuration, artifacts, services, or infrastructure to a known, authorized, coherent, and usable state after loss, corruption, migration, or recovery action.

REQUIRED OUTCOME

A restore model with source, point in time, target, authority, integrity, dependency, sequencing, validation, reconciliation, rollback, communication, and evidence.

WHY THIS MATTERS

Restore can overwrite good state, reintroduce vulnerabilities, violate retention, or produce a technically available but semantically inconsistent system. A successful copy is not a restored outcome.

APPLICABILITY

Apply when backup, archive, snapshot, replica, prior artifact, or known-good state is used to return a system, data, or service to operation.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no state, data, service, or recovery path requires restoration and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use backup, storage, database, data, identity, security, continuity, DR, failover, change, validation, and operations baselines.

REQUIRED INPUTS

Source copy, point in time, target, authority, keys, dependencies, sequence, integrity, validation, reconciliation, rollback, and communication.

HUMAN RESPONSIBILITIES AND AUTHORITY

Recovery owners approve source, target, data loss, overwrite, access, sequence, communication, and return to service. Humans judge semantic and user impact.

AI RESPONSIBILITIES AND BOUNDARIES

AI may inspect authorized restore metadata, compare state, draft runbooks, and analyze results. It must not restore, overwrite, delete, expose data, or declare success without authority.

APPLICABLE EXISTING AGENT ROLES

Restore Engineer, Data Engineer, Database Administrator, Storage Administrator, DR Planner, Security Auditor, Tester, Verifier, and Incident Commander.

REQUIRED ACTIVITIES

Define recovery objective; select and verify source; prepare target; protect current state; restore in order; validate integrity and semantics; reconcile; communicate; monitor; roll back or retry; and record.

HUMAN-AI INTERACTION REQUIREMENTS

Show source, point in time, target, authority, data, keys, sequence, evidence, impact, and limitation. Require human approval for restore, overwrite, deletion, data exposure, and return to service.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use isolated restore, immutable source, key availability, dependency ordering, checkpoint, reconciliation, transaction, validation, rollback, and audit.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show restore phase, source, target, data point, integrity, dependencies, evidence freshness, owner, and authorized start, pause, rollback, reconcile, communicate, and accept controls.

REQUIRED OUTPUTS AND ARTIFACTS

Restore charter; source and target record; execution and integrity; validation and reconciliation; rollback and communication; decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify source, version, point in time, target, method, authority, time, result, integrity, reconciliation, actor, and limitation. Copy completion proves neither usable state nor correctness.

VERIFICATION AND ACCEPTANCE TESTS

Verify source, integrity, keys, target isolation, sequence, dependencies, data semantics, permissions, reconciliation, rollback, RTO/RPO, and user outcome.

FAILURE STATES AND ANTI-PATTERNS

Restore over current state, wrong point in time, key unavailable, dependency omitted, data integrity assumed, backup copy accepted without test, and return before validation.

RECOVERY AND RECONSIDERATION

Stop, preserve both states, roll back or choose alternate source, reconcile, communicate, and reassess data and user impact. Reopen on source, target, key, dependency, or validation change.

COMPLETION CRITERIA

Source, point in time, target, authority, integrity, sequence, validation, reconciliation, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve restore logs, source, target, checks, reconciliations, incidents, decisions, evidence, exceptions, and open gaps. Hand off to Backup, DR, Failover, Data, Database, Storage, Security, and Operations.

RELATED CODEX SECTIONS

328 Backup; 326-327 Continuity and Disaster Recovery; 330 Failover; 307 Data Operations; 285-286 Database and Storage; 391-395 Data Lifecycle.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, restore, data, storage, continuity, and operations context, and the integration roadmap. Exact sources, targets, keys, sequence, validation, RTO/RPO, and acceptance authority remain unresolved.

330

SECTION 330 - FAILOVER ENGINEERING

CODEX/03_SECTIONS/330_FAILOVER_ENGINEERING.txt

SHA-256 34361c681c4f75be7a141059d261209463dc4865c71035a6e0097ce48d403232

4,730 BYTES / PRESERVED SOURCE IDENTITY

SECTION 330 — Failover Engineering

STATUS: CONTENT-COMPLETE

PURPOSE

Transfer service, traffic, workload, or control from a failed or degraded component to an alternate while preserving safety, integrity, authority, and recovery evidence.

REQUIRED OUTCOME

A failover model with trigger, authority, primary and alternate, data state, synchronization, health, sequencing, split-brain control, return, rollback, and evidence.

WHY THIS MATTERS

Failover can duplicate authority, lose data, route users to an unready target, or make recovery harder. Redundancy without tested control and reconciliation is not resilience.

APPLICABILITY

Apply when a service, network, data store, region, provider, device, or control plane has an alternate path or instance.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no alternate, continuity, dependency, or availability claim changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use availability, reliability, continuity, DR, backup, restore, network, database, identity, application, and operational readiness baselines.

REQUIRED INPUTS

Primary, alternate, trigger, authority, data sync, health, traffic, dependencies, sequence, split-brain prevention, return, and acceptance.

HUMAN RESPONSIBILITIES AND AUTHORITY

Owners approve trigger, failover, data loss, priority, communication, return, and residual risk. Humans can stop unsafe or duplicate operation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may analyze health and sync metadata, draft failover plans, and recommend action. It must not fail over, switch traffic, promote data, or declare alternate ready without authority.

APPLICABLE EXISTING AGENT ROLES

Failover Engineer, SRE, DR Planner, Network Engineer, Database Administrator, Operations Engineer, Tester, Verifier, Security Auditor, and Incident Commander.

REQUIRED ACTIVITIES

Define trigger and authority; validate alternate; maintain synchronization; test; detect failure; declare; sequence failover; monitor; reconcile; return or retire primary; and record.

HUMAN-AI INTERACTION REQUIREMENTS

Show primary, alternate, trigger, health, data point, authority, traffic, evidence, and limitation. Require human approval for declaration, promotion, data, routing, and return.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Use independent failure domains, fencing, leader election, quorum, data consistency, health, traffic control, identity, observability, rollback, and split-brain prevention.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show primary and alternate, sync, health, mode, traffic, data loss, evidence freshness, owner, and authorized declare, fence, fail over, reconcile, return, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Failover charter; primary and alternate; trigger and authority; sync and health; exercise and runbook; reconciliation, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify primary, alternate, trigger, version, data point, authority, method, time, result, traffic, integrity, and limitation. A healthy standby proves neither safe promotion nor current data.

VERIFICATION AND ACCEPTANCE TESTS

Verify trigger, fencing, sync, health, identity, traffic, data integrity, failover, user outcome, return, rollback, and operator response. Test split brain and stale standby.

FAILURE STATES AND ANTI-PATTERNS

Both sides active, standby untested, stale data, failover loop, traffic switched without health, no fence, return without reconciliation, and redundancy treated as recovery.

RECOVERY AND RECONSIDERATION

Fence unsafe primary, preserve evidence, fail over by priority, reconcile, communicate, restore or replace, and reassess data and service impact. Reopen on dependency, sync, trigger, topology, or recovery change.

COMPLETION CRITERIA

Primary, alternate, trigger, authority, sync, fencing, health, traffic, recovery, evidence, owners, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve failover records, sync, health, incidents, exercises, decisions, evidence, exceptions, and open gaps. Hand off to DR, Restore, Backup, Network, Database, SRE, and Operations.

RELATED CODEX SECTIONS

322 Availability; 324 Reliability; 326-329 Continuity, DR, Backup, and Restore; 301 SRE; 305 Network; 285 Database; 331 Operational Readiness.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, failover, continuity, data, network, SRE, and operations context, and the integration roadmap. Exact primary, alternate, trigger, sync, RTO/RPO, authority, and acceptance evidence remain unresolved.

331

SECTION 331 - OPERATIONAL READINESS

CODEX/03_SECTIONS/331_OPERATIONAL_READINESS.txt

SHA-256 c7082267c77e3727b7f4959008819a0525c590790a70e26b1a7125494fa0622f

5,005 BYTES / PRESERVED SOURCE IDENTITY

SECTION 331 — Operational Readiness

STATUS: CONTENT-COMPLETE

PURPOSE

Determine whether a service, system, product, process, or change can be safely operated, supported, monitored, secured, recovered, and owned in its declared environment.

REQUIRED OUTCOME

An operational-readiness record with owner, scope, modes, runbooks, staffing, access, observability, support, security, capacity, continuity, recovery, evidence, gaps, and explicit decision.

WHY THIS MATTERS

Deployment or acceptance can occur before operations can safely sustain the outcome. Readiness is a decision with limitations, not a checklist or a green dashboard.

APPLICABILITY

Apply before entering a new operational mode, production, field use, support commitment, or material transition.

NON-APPLICABILITY CONDITIONS

Record non-applicability when no operational ownership, support, mode, risk, user, dependency, or recovery condition changes and an approved exact baseline covers it.

PREREQUISITES AND DEPENDENCIES

Use operational architecture, acceptance, release, deployment, security, privacy, support, SRE, observability, capacity, continuity, recovery, and training baselines.

REQUIRED INPUTS

Service, environment, owner, users, modes, runbooks, staffing, access, alerts, logs, traces, metrics, support, risks, capacity, backup, recovery, training, and approvals.

HUMAN RESPONSIBILITIES AND AUTHORITY

Accountable owners decide ready, conditionally ready, hold, or reject. Operators and support owners confirm capability; security, privacy, safety, accessibility, and continuity authorities approve affected conditions.

AI RESPONSIBILITIES AND BOUNDARIES

AI may assemble evidence, identify missing readiness items, draft runbooks and exercises, and summarize risk. It must not declare readiness, accept residual risk, or approve operation without human authority.

APPLICABLE EXISTING AGENT ROLES

Operational Readiness Reviewer, Operations Engineer, SRE, Release Manager, Support Reviewer, Security Auditor, Tester, Verifier, Coordinator, and Governance Reviewer.

REQUIRED ACTIVITIES

Define scope and modes; confirm owners and access; validate runbooks, monitoring, support, security, capacity, backup, recovery, training, documentation, and communication; exercise; record gaps; decide; and schedule review.

HUMAN-AI INTERACTION REQUIREMENTS

Show readiness condition, evidence, owner, gap, risk, environment, support, authority, and decision. Require human sign-off and preserve conditional or rejected status.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Provide operational interfaces, health, logs, traces, metrics, access, safe mode, support, rollback, backup, restore, runbooks, and owner control before operation.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show readiness state, owner, environment, gaps, evidence freshness, incidents, support, training, recovery, and authorized approve, condition, hold, reject, remediate, and reopen controls.

REQUIRED OUTPUTS AND ARTIFACTS

Readiness charter; owner and mode; runbooks; support and training; security and privacy; capacity, observability, continuity, recovery, decision, evidence, and gap records.

EVIDENCE REQUIREMENTS

Identify condition, system, version, environment, method, result, owner, reviewer, date, validity, gap, and limitation. Checklist completion proves neither operational effectiveness nor adoption.

VERIFICATION AND ACCEPTANCE TESTS

Verify ownership, access, runbook, monitoring, alert, logs, traces, metrics, support, security, capacity, backup, restore, failover, training, communication, and user recovery.

FAILURE STATES AND ANTI-PATTERNS

Readiness by checklist, owner missing, support unprepared, stale evidence, no recovery, undocumented exception, conditional ready without expiry, and deployment treated as readiness.

RECOVERY AND RECONSIDERATION

Hold or narrow operation, preserve evidence, remediate gaps, support users, and reconvene decision authority. Reopen on version, environment, owner, mode, dependency, incident, or evidence change.

COMPLETION CRITERIA

Scope, owner, modes, evidence, runbooks, support, security, observability, capacity, recovery, gaps, decision, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve readiness package, decisions, gaps, approvals, runbooks, training, evidence, exceptions, and open actions. Hand off to Operations, SRE, Support, Security, Continuity, Release, and Governance.

RELATED CODEX SECTIONS

029 Operational Architecture; 237 Acceptance; 288-290 Release, Change, Deployment; 301-330 Operations and Recovery; 332 Command and Control.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, operational, acceptance, support, reliability, and owner-control context, and the integration roadmap. Exact service, owner, environment, evidence, staffing, training, recovery, and acceptance authority remain unresolved.

332

SECTION 332 - COMMAND AND CONTROL

CODEX/03_SECTIONS/332_COMMAND_AND_CONTROL.txt

SHA-256 04d66b65994d66c85a0d514c245797b8245786a05df8e16aa1860addcd973a16

5,199 BYTES / PRESERVED SOURCE IDENTITY

SECTION 332 — Command and Control

STATUS: CONTENT-COMPLETE

PURPOSE

Define who or what may observe, decide, direct, authorize, pause, stop, escalate, recover, and retire consequential operations across normal and emergency modes.

REQUIRED OUTCOME

A command-and-control model with authority, identity, scope, mode, decision rights, delegation, communication, status, override, evidence, audit, safety, and recovery.

WHY THIS MATTERS

Systems fail when authority is ambiguous, duplicated, unavailable, or silently transferred to automation. Command must be visible, bounded, revocable, and accountable under pressure.

APPLICABILITY

Apply when people, automation, AI, services, devices, or organizations can direct consequential operation, resource, safety, security, user, or recovery action.

NON-APPLICABILITY CONDITIONS

Record non-applicability only when no consequential decision, control, delegation, mode, escalation, or emergency action exists and an approved exact baseline covers the scope.

PREREQUISITES AND DEPENDENCIES

Use authority, identity, operational, incident, continuity, safety, security, service, change, communication, and governance baselines.

REQUIRED INPUTS

Actors, roles, authority, resources, modes, decisions, delegation, escalation, communication, status, evidence, override, stop, and recovery.

HUMAN RESPONSIBILITIES AND AUTHORITY

Accountable humans define authority, delegation, limits, escalation, emergency, stop, communication, and acceptance. Humans remain responsible for protected outcomes and can revoke automation.

AI RESPONSIBILITIES AND BOUNDARIES

AI may surface state, summarize options, and recommend bounded action. It must identify its authority and never self-escalate, impersonate, hide dissent, execute consequential action, or block human override.

APPLICABLE EXISTING AGENT ROLES

Coordinator, Incident Commander, Governance Reviewer, Architect, Operations Engineer, Security Auditor, Verifier, Planner, Critic, and Documentation Writer.

REQUIRED ACTIVITIES

Map authority and actors; define modes and decisions; bind identity; establish delegation and expiry; provide command channel; communicate; record; exercise stop and override; recover; audit; and review.

HUMAN-AI INTERACTION REQUIREMENTS

Show actor, authority, scope, mode, evidence, recommendation, action, consequence, revocation, and limitation. Require human confirmation for protected, irreversible, emergency, safety, rights, and external actions.

ARCHITECTURE AND SYSTEM IMPLICATIONS

Separate observe, advise, decide, authorize, execute, audit, and recover. Use least privilege, dual control, expiry, quorum where needed, independent stop, resilient communication, and immutable decision record.

OWNER COCKPIT AND USER-CONTROL REQUIREMENTS

Show current commander, mode, authority, active delegations, decisions, evidence freshness, actions, conflicts, and authorized pause, stop, revoke, escalate, override, recover, and retire controls.

REQUIRED OUTPUTS AND ARTIFACTS

Command charter; authority and delegation matrix; mode and decision model; communication and escalation; stop and override; audit, decision, evidence, exception, and continuity records.

EVIDENCE REQUIREMENTS

Identify actor, identity, authority, scope, mode, decision, source, time, action, result, reviewer, and limitation. Command presence proves neither legitimacy nor correctness without evidence and scope.

VERIFICATION AND ACCEPTANCE TESTS

Verify identity, authority, delegation, expiry, conflict, escalation, stop, override, communication, audit, unavailable commander, automation failure, and recovery. Test unauthorized and ambiguous command.

FAILURE STATES AND ANTI-PATTERNS

Shared command, hidden delegation, unrevocable automation, no stop path, authority by chat alone, multiple conflicting commanders, emergency exception permanent, and status without decision record.

RECOVERY AND RECONSIDERATION

Freeze unsafe action, revoke authority, establish accountable commander, preserve records, communicate, restore safe mode, and reassess decisions. Reopen on authority, actor, mode, system, incident, or policy change.

COMPLETION CRITERIA

Actors, authority, modes, decisions, delegation, communication, stop, override, evidence, audit, recovery, limitations, and reopening conditions are explicit. Status remains CONTENT-COMPLETE pending owner acceptance.

HANDOFF AND CONTINUITY REQUIREMENTS

Preserve authority matrix, delegations, command timeline, decisions, communications, incidents, evidence, exceptions, and open gaps. Hand off to Operations, Incident, Continuity, Safety, Security, Governance, and Retirement.

RELATED CODEX SECTIONS

003 Authority and Delegation; 029 Operational Architecture; 214 Incident Response; 319 Incident Management; 326-331 Continuity, DR, Restore, Failover, and Readiness; 402-406 Review and Retirement.

SOURCE BASIS AND UNRESOLVED SOURCE GAPS

Based on the Canonical Section Contract, shared governance, authority, operations, incident, continuity, safety, and human-control context, and the integration roadmap. Exact actors, authorities, modes, emergency, delegation, communication, and acceptance authority remain unresolved.